

3. Linear Regression

3.1 Simple linear regression

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3.3 Least squares estimation (LSE)

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3.4 Statistical properties of LSE

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3.0 Introduction

Fundamental statistical models. (supervised learning)

Covering one-sample, two-sample, multiple sample problems.

Most illustrative on various important issues: fitting, prediction, model checking, ...

In-depth understanding of linear model helps learning further topics.

3.0 Introduction

Formulation

General linear regression model

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \epsilon_i, \quad i = 1, \dots, n$$

where $y_i, x_i = (x_{i1}, \dots, x_{ip})$ are the i -th observation of the response and covariates.

Responses are sometimes called dependent variables or outputs;

covariates called independent variables or inputs or regressors.

Goal

obtain the parameter estimation and making prediction of any responses on given covariates.

3.0 Introduction

Example: advertising data

The data contains 200 observations.

Sample size: $n = 200$.

Sales: $y_i, i = 1, \dots, n$.

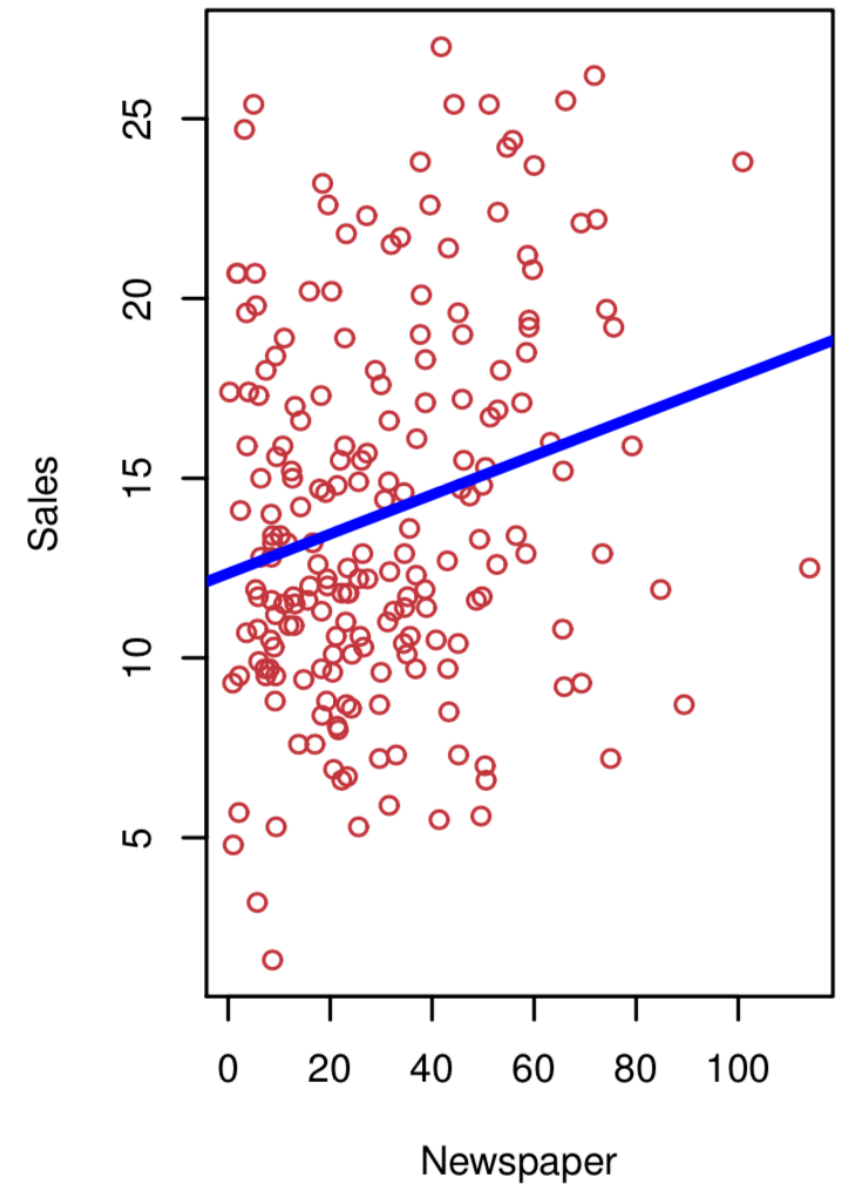
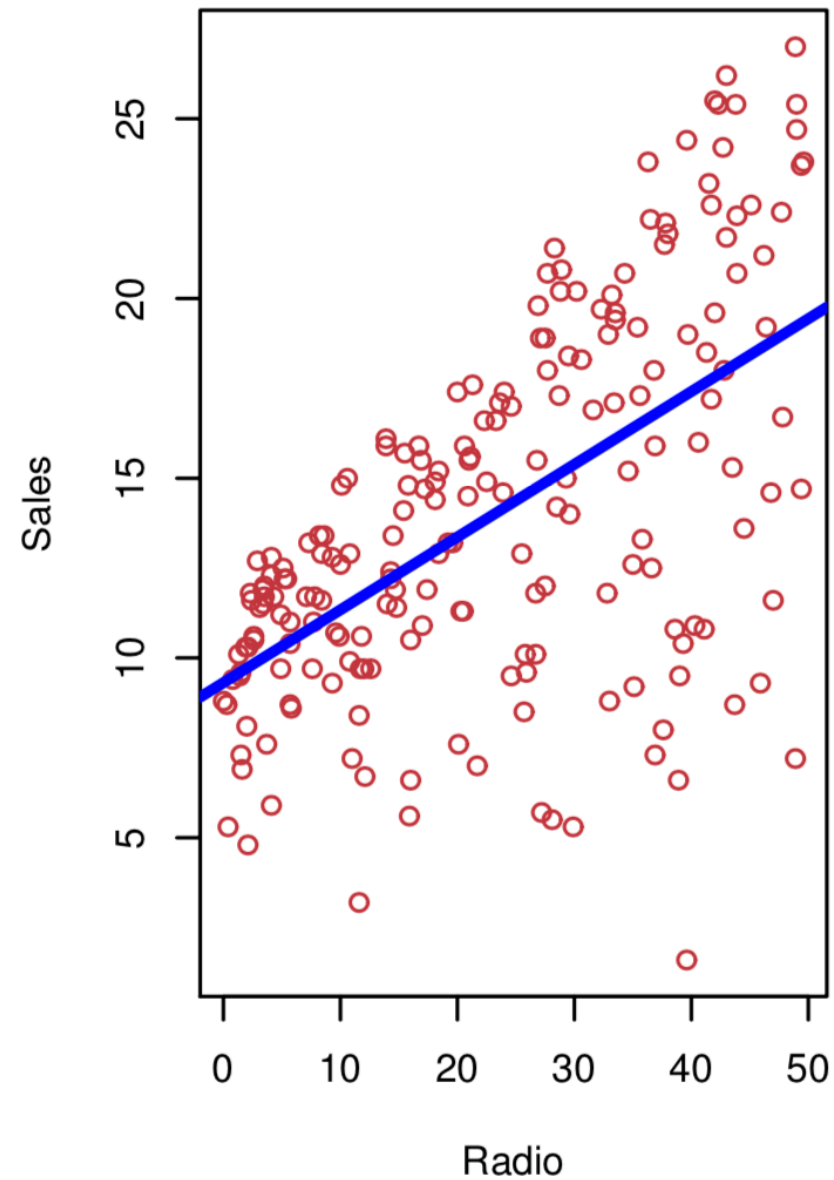
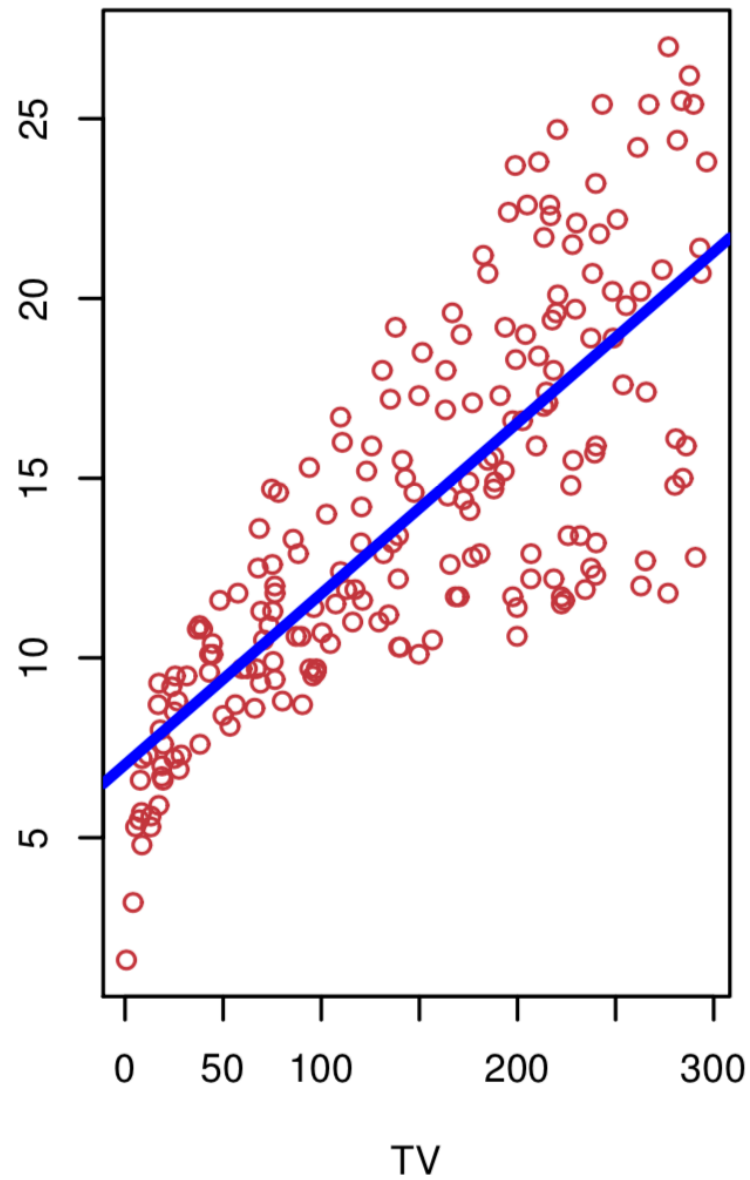
TV (budgets): $x_{i1}, i = 1, \dots, n$.

Radio (budgets): $x_{i2}, i = 1, \dots, n$.

Newspaper (budgets): $x_{i3}, i = 1, \dots, n$.

3.0 Introduction

Example: advertising data



3.1 Simple Linear Regression

If $p = 1$, i.e. only one covariate, we call it a simple linear regression model.

Simple linear regression

$$y_i = \beta_0 + \beta_1 x_{i1} + \varepsilon_i, \quad i = 1, \dots, n$$

Minimize the sum of squares of err: $\min_{a,b} \sum_{i=1}^n (y_i - a - bx_i)^2$

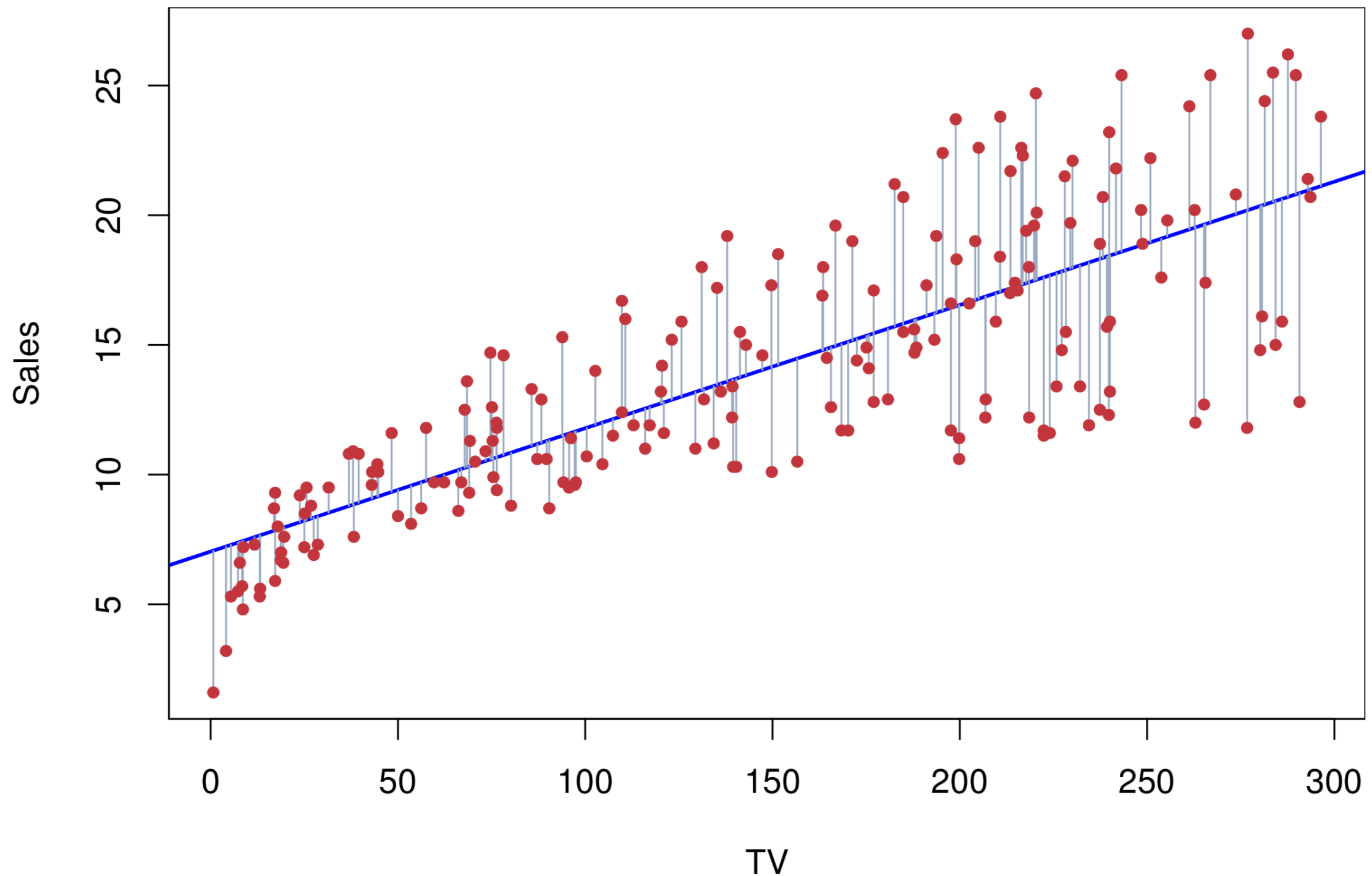
The estimator is
$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

we write x_i in short for x_{i1}

3.1 Simple Linear Regression

Example: advertising data



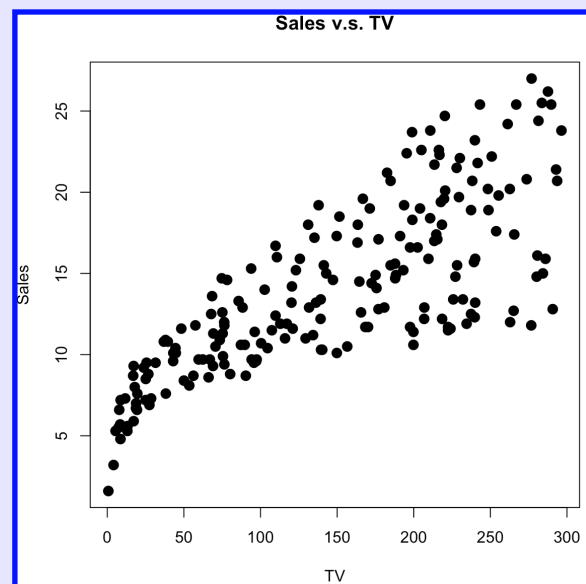
3.1 Simple Linear Regression

Example: advertising data by R

```
> adv<-read.csv('Datasets/Advertising.csv',header=TRUE)
> adv[1:10,]
```

	TV	Radio	Newspaper	Sales
1	230.1	37.8	69.2	22.1
2	44.5	39.3	45.1	10.4
3	17.2	45.9	69.3	9.3
4	151.5	41.3	58.5	18.5
5	180.8	10.8	58.4	12.9
6	8.7	48.9	75.0	7.2
7	57.5	32.8	23.5	11.8
8	120.2	19.6	11.6	13.2
9	8.6	2.1	1.0	4.8
10	199.8	2.6	21.2	10.6

```
# scatter plot to check the linearity
plot(Sales ~ TV, data = adv,
     xlab = "TV",
     ylab = "Sales",
     main = "Sales v.s. TV",
     pch = 20,
     cex = 2,
     col = "black")
```



```
> mod1<-lm(Sales~TV,data=adv)
> summary(mod1)
```

Call:
lm(formula = Sales ~ TV, data = adv)

Residuals:

Min	1Q	Median	3Q	Max
-8.3860	-1.9545	-0.1913	2.0671	7.2124

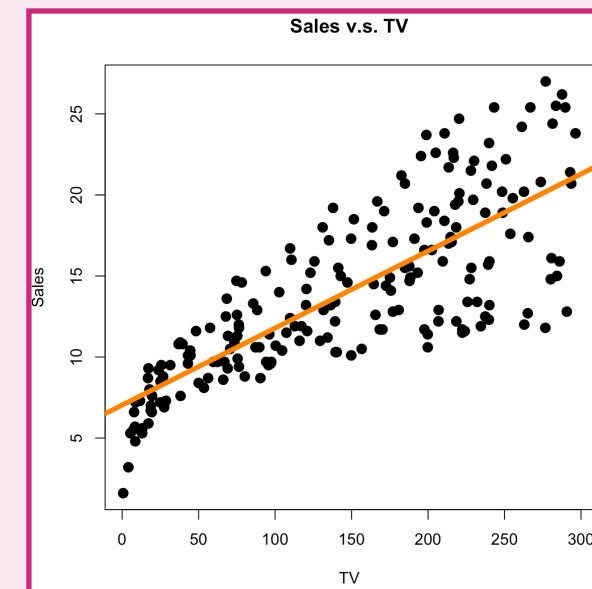
Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	7.032594	0.457843	15.36	<2e-16 ***
TV	0.047537	0.002691	17.67	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.259 on 198 degrees of freedom
Multiple R-squared: 0.6119, Adjusted R-squared: 0.6099
F-statistic: 312.1 on 1 and 198 DF, p-value: < 2.2e-16

```
abline(mod1, lwd = 5, col = "darkorange")
```



3.1 Simple Linear Regression

Statistical Inference

How accurate are $\hat{\beta}_0$ and $\hat{\beta}_1$?

The distributions of $\hat{\beta}_0$ and $\hat{\beta}_1$?

Goal

To quantify the uncertainty of $\hat{\beta}_0$ and $\hat{\beta}_1$?

Assumption

$$\varepsilon_i \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2)$$

The random error terms are independent and have identical normal distributions.

3.1 Simple Linear Regression

Statistical Inference

Theorem

$$\frac{\hat{\beta}_1 - \beta_1}{s \sqrt{1 / \sum_{i=1}^n (x_i - \bar{x})^2}} \sim t_{n-2} \quad \frac{\hat{\beta}_0 - \beta_0}{s \sqrt{1/n + \bar{x}^2 / \sum_{i=1}^n (x_i - \bar{x})^2}} \sim t_{n-2}$$

$\text{s.e.}(\hat{\beta}_1)$
 $\text{s.e.}(\hat{\beta}_0)$

called standard error

Student t distribution

 t_{n-2}
https://en.wikipedia.org/wiki/Student%27s_t-distribution

Fitted value

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

Residual

$$e_i = y_i - \hat{y}_i$$

Residual sum of squares

$$s^2 = \text{RSS} / (n - 2)$$

$$\text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

3.1 Simple Linear Regression

Statistical Inference

Example (from ISLR) The advertising data: coefficients of the LSE for the regression on number of units sold on TV advertising budget. An increase of \$1000 in the TV advertising budget would cause an increase of sales of about 50 units.

		Coefficient	Std.error	t-statistic	p-value
$\hat{\beta}_0$	Intercept	7.0325	0.4578	15.36	< 0.0001
$\hat{\beta}_1$	TV	0.0475	0.0027	17.67	< 0.0001

```
> mod1<-lm(Sales~TV,data=adv)
> summary(mod1)

Call:
lm(formula = Sales ~ TV, data = adv)

Residuals:
    Min       1Q   Median       3Q      Max
-8.3860 -1.9545 -0.1913  2.0671  7.2124

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  7.032594   0.457843   15.36  <2e-16 ***
TV           0.047537   0.002691   17.67  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.259 on 198 degrees of freedom
Multiple R-squared:  0.6119,    Adjusted R-squared:  0.6099
F-statistic: 312.1 on 1 and 198 DF,  p-value: < 2.2e-16
```

Smaller p-value indicates that this coefficient is more likely non-zero.

$$t\text{-statistic} = \frac{\hat{\beta}_1}{\text{s.e.}(\hat{\beta}_1)}$$

3.2 Multiple Linear Regression

General linear regression model

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \epsilon_i, \quad i = 1, \dots, n$$

where $y_i, x_i = (x_{i1}, \dots, x_{ip})$ are the i -th observation of the response and covariates.

Example

Consider the advertising data with three covariates: TV, radio, newspapers.

The number of covariates $p=3$

The linear regression model is

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + \epsilon_i, \quad i = 1, \dots, n$$

3.2 Multiple Linear Regression

Example: advertising data

Minimizing the sum of squares of error:

$$\sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_{i1} - \beta_2 x_{i2} - \beta_3 x_{i3})^2 = \sum_{i=1}^n (y_i - \beta^T x_i)^2$$

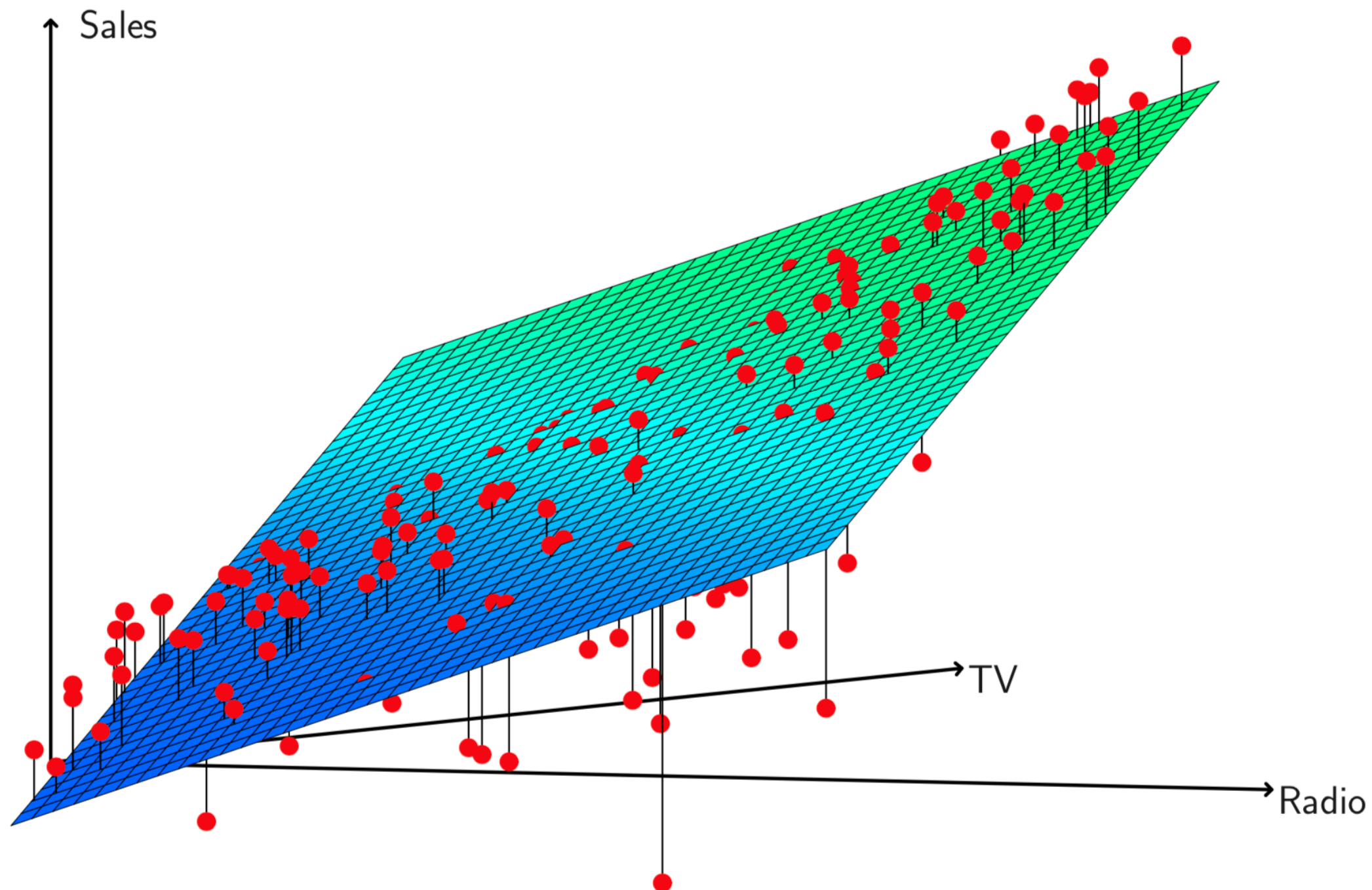
Derivative (formally called gradient) of $\sum_{i=1}^n (y_i - \beta^T x_i)^2$

$$2 \sum_{i=1}^n (\beta^T x_i - y_i) x_i^T = 2\beta^T \left(\sum_{i=1}^n x_i x_i^T \right) - 2 \sum_{i=1}^n y_i x_i^T$$

Setting the gradient zero and solve β

3.2 Multiple Linear Regression

Example: advertising data



3.2 Multiple Linear Regression

Example: advertising data

Inference provided by R The advertising data: coefficients of the LSE for the regression on number of units sold on TV, radio and newspaper advertising budgets.

	Coefficient	Std.error	t-statistic	p-value
Intercept	2.939	0.3119	9.42	< 0.0001
TV	0.046	0.0014	32.81	< 0.0001
radio	0.189	0.0086	21.89	< 0.0001
newspaper	-0.001	0.0059	-0.18	0.8599

3.2 Multiple Linear Regression

Example: advertising data by R

```
> mod2<-lm(Sales~.,data=adv)
> summary(mod2)

Call:
lm(formula = Sales ~ ., data = adv)

Residuals:
    Min       1Q   Median       3Q      Max
-8.8277 -0.8908  0.2418  1.1893  2.8292

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.938889    0.311908   9.422  <2e-16 ***
TV            0.045765    0.001395  32.809  <2e-16 ***
Radio        0.188530    0.008611  21.893  <2e-16 ***
Newspaper   -0.001037    0.005871  -0.177    0.86
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.686 on 196 degrees of freedom
Multiple R-squared:  0.8972,    Adjusted R-squared:  0.8956
F-statistic: 570.3 on 3 and 196 DF,  p-value: < 2.2e-16
```

```
> names(mod2)
[1] "coefficients" "residuals"    "effects"      "rank"         "fitted.values" "assign"       "qr"           "df.residual"  "xlevels"      "call"         "terms"
```

```
> round(mod2$residuals,digits=4)
     1      2      3      4      5      6      7      8      9     10     11     12     13     14     15     16     17     18     19     20     21     22
1.5760 -1.9379 -3.0077  0.9022 -0.2887 -5.2783  0.0702  1.0770  1.0727 -1.9508  1.5677  0.1149 -1.3771  0.8737  0.5656  1.5807 -0.3237  1.1750  1.3483  0.4339 -0.1008 -2.2405
23      24      25      26      27      28      29      30      31      32      33      34      35      36      37      38      39      40      41      42      43      44
-0.8892 -1.0459  1.5535 -3.6100  0.0105 -1.1517 -0.5105  1.3560 -0.2339  0.5539  1.9611 -1.4643  1.9252 -4.2068  1.9941 -0.9235  0.1913  1.0524  0.2223 -0.1960 -0.8958 -1.0639
45      46      47      48      49      50      51      52      53      54      55      56      57      58      59      60      61      62      63      64      65      66
-0.3879 -0.2615  1.7266  1.4774 -1.4636  1.5318 -1.2312  1.3602  1.9370  1.2553 -0.1744  2.4074 -3.0277  0.4254  1.9019  0.2665  2.3578  1.3093 -1.0843  0.7893  1.0223  1.4510
67      68      69      70      71      72      73      74      75      76      77      78      79      80      81      82      83      84      85      86      87      88
0.4840  1.3630 -0.0766  1.1911  0.5205  1.7731 -1.5668  1.0970 -0.3293 -3.1583  2.4224  0.3881 -3.5133  1.3247  0.3541 -2.3479  1.1216 -0.8218  0.9186  0.0186  0.4013  0.4062
89      90      91      92      93      94      95      96      97      98      99     100     101     102     103     104     105     106     107     108     109     110
1.1887 -0.2223  1.2008  2.8037  0.2436  0.9724  1.0179  0.5851 -0.9357  0.1629  1.2814  0.2596 -2.1760  0.5575 -2.8441 -0.0622  0.3989  1.2636  1.0740  1.5915  1.7127  0.1071
111     112     113     114     115     116     117     118     119     120     121     122     123     124     125     126     127     128     129     130     131     132
-1.3599  0.6597  0.2194 -0.5038 -0.7051 -0.3197  0.2213  2.8292  0.3339 -0.2201  1.0899 -0.8381 -2.0265  0.1172  0.2456  1.4727 -3.9772  2.2003  2.4445  1.8159 -8.8277 -2.8778
133     134     135     136     137     138     139     140     141     142     143     144     145     146     147     148     149     150     151     152     153     154
-2.7492  0.3331 -1.0368 -2.4014 -1.9535 -0.0513 -0.1684  1.0245  1.4104  0.8010  0.8501  1.6352  1.3087  0.5915 -2.0942  2.1391 -1.3634  0.2727 -2.2672  1.5905  0.2400  0.7761
155     156     157     158     159     160     161     162     163     164     165     166     167     168     169     170     171     172     173     174     175     176
0.0984 -2.1076 -0.0849  0.0857 -3.0842  0.5009  0.1862 -0.2591 -0.0468  0.6484  0.8317 -2.3237 -2.8244 -1.1632 -0.0861 -2.9416  1.0050  0.1417 -0.0077 -0.2709 -2.2444  2.2131
177     178     179     180     181     182     183     184     185     186     187     188     189     190     191     192     193     194     195     196     197     198
0.2206 -0.4620 -4.2110  0.2154 -0.0872 -1.7281  2.1453  2.0669 -0.9385  1.7970  0.6086  0.2236 -2.7443  0.6484 -1.6892  1.4760  1.4338  1.1130  0.8047  2.2297  1.5347  0.0141
199     200
1.7327 -1.7732
```

3.2 Multiple Linear Regression

Matrix Notation

In this chapter, we write

$$\mathbf{X} = \begin{pmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1p} \\ 1 & x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{np} \end{pmatrix} \quad \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \cdots \\ \beta_p \end{pmatrix}$$

$$= \begin{pmatrix} \mathbf{1} & \mathbf{x}_1 & \mathbf{x}_2 & \cdots & \mathbf{x}_p \end{pmatrix}$$

Then

$$\mathbf{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \quad \beta = \begin{pmatrix} \beta_0 \\ \vdots \\ \beta_p \end{pmatrix}, \quad \epsilon = \begin{pmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{pmatrix}, \quad \mathbf{x}_j = \begin{pmatrix} x_{1j} \\ \vdots \\ x_{nj} \end{pmatrix}$$

3.3 Least Squares Estimation

The least squares criterion is to minimize the sum of squares

$$\sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_{i1} - \cdots - \beta_p x_{ip})^2$$

Using matrix algebra, it is equivalently written as

$$\|\mathbf{y} - \mathbf{X}\beta\|^2 = (\mathbf{y} - \mathbf{X}\beta)^T (\mathbf{y} - \mathbf{X}\beta)$$

By the first order optimality condition (zeroing the gradient), the LSE of β is

$$\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

Fitted value

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\beta}$$

Residual

$$\hat{\varepsilon} = \mathbf{y} - \hat{\mathbf{y}}$$

3.3 Least Squares Estimation

Orthogonality of residual and covariates



The residual $\hat{\epsilon}$ is orthogonal to all columns of $\mathbf{X}$: $\mathbf{X}^T \hat{\epsilon} = \mathbf{0}$

$$\begin{aligned}\mathbf{X}^T \hat{\epsilon} &= \mathbf{X}^T \mathbf{y} - \mathbf{X}^T \mathbf{X} \hat{\beta} \\ &= \mathbf{X}^T \mathbf{y} - \mathbf{X}^T \mathbf{X} (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} = \mathbf{0}.\end{aligned}$$

The residual vector $\hat{\epsilon}$ is orthogonal to the hyperplane formed by vectors $\mathbf{1}, \mathbf{x}_1, \dots, \mathbf{x}_p$ in n dimensional real space.

Proof of LSE

For any vector $v \in \mathbb{R}^{p+1}$

$$\begin{aligned}\|\mathbf{y} - \mathbf{X}v\|^2 &= \|\mathbf{y} - \mathbf{X}\hat{\beta} - \mathbf{X}(v - \hat{\beta})\|^2 \\ &= \|\mathbf{y} - \mathbf{X}\hat{\beta}\|^2 + \|\mathbf{X}(v - \hat{\beta})\|^2 \geq \|\mathbf{y} - \mathbf{X}\hat{\beta}\|^2\end{aligned}$$

3.3 Least Squares Estimation

Orthogonality of residual and fitted values



The fitted value $\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$, which, also as a vector in n dimensional real space, is a linear combination of the vectors $\mathbf{1}, \mathbf{x}_1, \dots, \mathbf{x}_p$, with the $p + 1$ linear combination coefficients being the components of $\hat{\boldsymbol{\beta}}$.

The fitted values are orthogonal to the residuals, i.e., $\hat{\mathbf{y}}$ is orthogonal to $\mathbf{y} - \hat{\mathbf{y}}$ or

$$\hat{\mathbf{y}}^T (\mathbf{y} - \hat{\mathbf{y}}) = 0.$$

This implies

$$\|\mathbf{y}\|^2 = \|\hat{\mathbf{y}}\|^2 + \|\mathbf{y} - \hat{\mathbf{y}}\|^2.$$

3.3 Least Squares Estimation

Projection matrix

$$\text{Let } \mathbf{H} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T.$$

This n by n matrix is called projection matrix or hat matrix.

It has the property that, for any vector, $\mathbf{b}$ in n dimensional real space $\mathbf{Hb}$ projects $\mathbf{b}$ onto the linear space formed by the columns of $\mathbf{X}$.

$\mathbf{Hb}$ is in this linear space formed by the columns of $\mathbf{X}$.

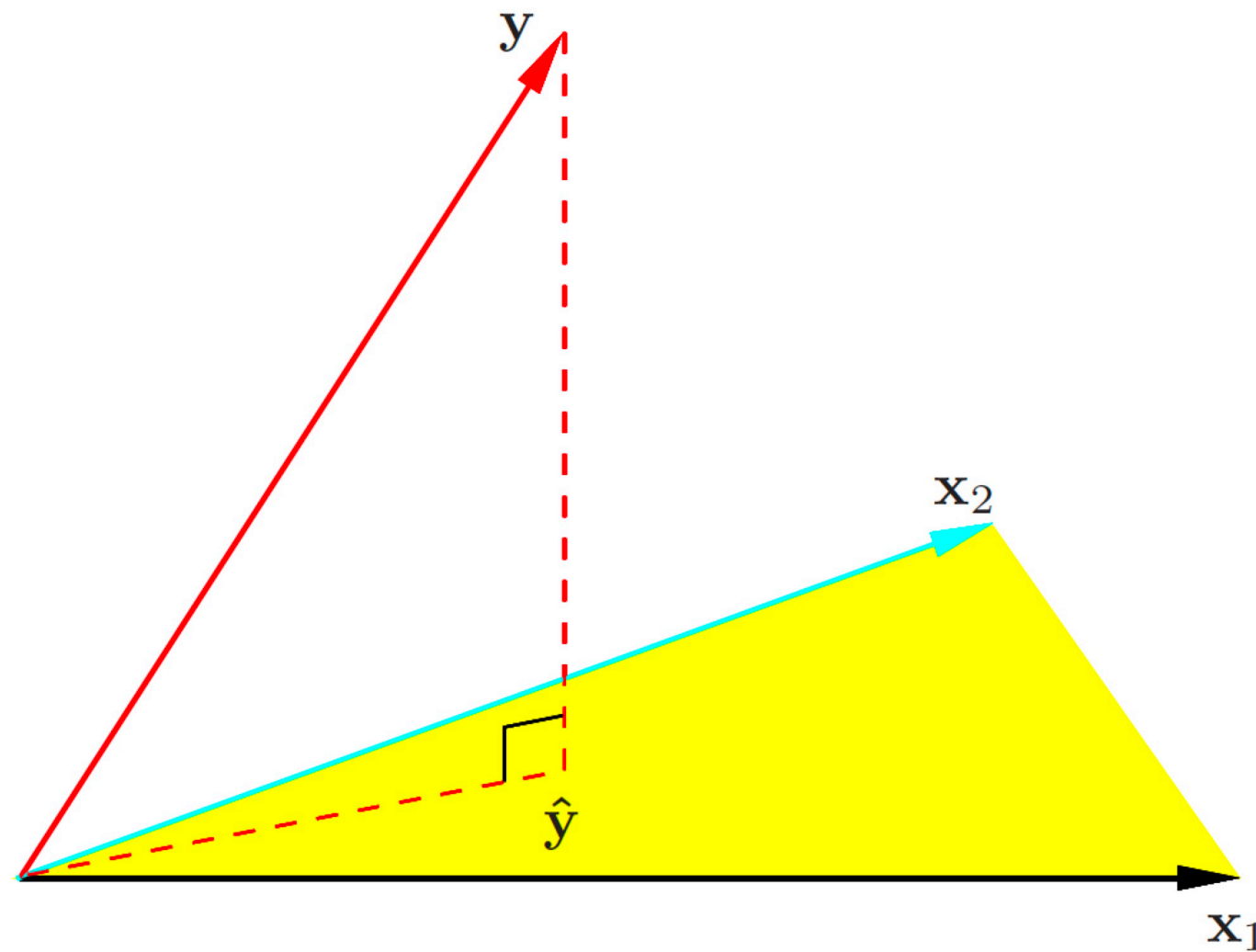
And $\mathbf{b} - \mathbf{Hb}$ is orthogonal to this space.

LSE $\hat{\mathbf{y}} = \mathbf{Hy}$

Residual $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{H})\mathbf{y}$

3.3 Least Squares Estimation

Projection matrix



The N -dimensional geometry of least squares regression with two predictors. The outcome vector y is orthogonally projected onto the hyperplane spanned by the input vectors x_1 and x_2 . The projection $\hat{y}$ represents the vector of the least squares predictions

3.3 Least Squares Estimation

Properties of projection matrix

projection matrix $\mathbf{H}$ is symmetric and idempotent; i.e., $\mathbf{H}^2 = \mathbf{H}$.

eigenvalues are either 1 or 0.

All eigenvectors associated with eigenvalue 1 form a space, say $\mathcal{L}_1$;

Those with eigenvalue 0 form the orthogonal space, $\mathcal{L}_0$, of $\mathcal{L}_1$.

Then $\mathbf{H}$ is the projection onto space $\mathcal{L}_1$ and $\mathbf{I} - \mathbf{H}$ is the projection onto $\mathcal{L}_0$, where $\mathbf{I}$ is the n by n identity matrix.

3.3 Least Squares Estimation

Singular Value Decomposition

Suppose, for convenience $n \geq p$, any matrix n by p matrix $\mathbf{X}$ can always be decomposed into

$$\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{V}^T$$

where $\mathbf{U}$ is $n \times p$ orthogonal matrix, $\mathbf{D}$ is a $p \times p$ diagonal matrix and $\mathbf{V}$ is $p \times p$ orthogonal matrix. In particular

$$\mathbf{X} = \mathbf{U}\mathbf{R},$$

where $\mathbf{R} = \mathbf{D}\mathbf{V}^T$.

Then, $\mathbf{H} = \mathbf{U}\mathbf{U}^T$ is the projection matrix on the column space of $\mathbf{X}$

3.4 Statistical Properties of LSE

Model assumptions

General linear regression model

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \epsilon_i, \quad i = 1, \dots, n$$

where $y_i, x_i = (x_{i1}, \dots, x_{ip})$ are the i -th observation of the response and covariates.

Assumptions

$$\epsilon_i \sim N(0, \sigma^2)$$

$(x_1, y_1), \dots, (x_n, y_n)$ are independent

3.4 Statistical Properties of LSE

Theorem of LSE

$$\hat{\beta} \sim N(\beta, \sigma^2(\mathbf{X}^T \mathbf{X})^{-1})$$

$$\frac{\hat{\beta}_j - \beta_j}{\underbrace{s \sqrt{c_{jj}}}_{\text{s.e.}(\hat{\beta}_j)}} \sim t_{n-p-1}$$

where $c_{00}, c_{11}, \dots, c_{pp}$ are the diagonal elements of $(\mathbf{X}^T \mathbf{X})^{-1}$

Theorem of Residual

$$\text{RSS} = \sum_{i=1}^n \hat{\epsilon}_i^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \sim \sigma^2 \chi_{n-p-1}^2 \quad \text{Chi-squared distribution}$$

$s^2 = \text{RSS}/(n - p - 1)$ unbiased estimate of σ^2

$\hat{\beta}$ and RSS are independent

$$\frac{(\hat{\beta} - \beta)^T (\mathbf{X}^T \mathbf{X}) (\hat{\beta} - \beta) / p}{s^2} \sim F_{p+1, n-p-1} \quad \text{F distribution}$$

3.4 Statistical Properties of LSE

simulation showing that $(\hat{\beta} - \beta) / \text{sd}(\beta)$ indeed follows t distribution.

```
import numpy as np
import matplotlib.pyplot as plt
import statsmodels.api as sm
from scipy import stats

# Parameters
n_simulations = 500
n_samples = 100
true_beta = np.array([1.5, -2.0, 3.0])
t_statistics = []

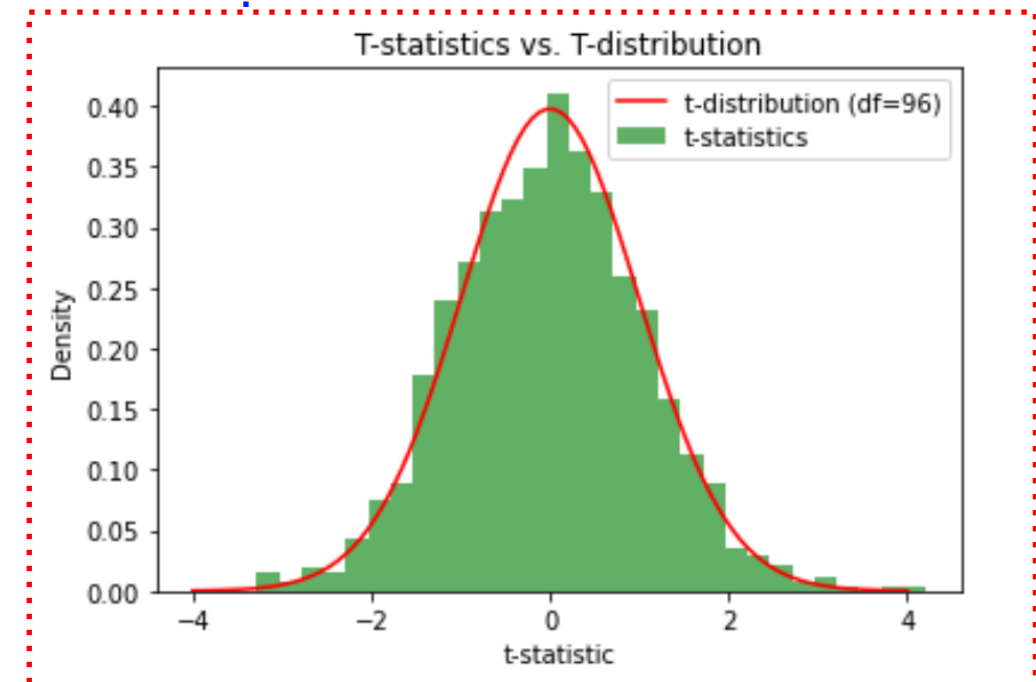
# Simulation loop
np.random.seed(0)
for _ in range(n_simulations):
    X = np.random.normal(size=(n_samples, 3))
    y = X @ true_beta + np.random.normal(scale=1.0, size=n_samples)

    # Fit the model
    X = sm.add_constant(X)
    model = sm.OLS(y, X).fit()

    # Calculate t-statistics
    estimated_beta = model.params[1:] # Exclude intercept
    se_beta = model.bse[1:] # Standard errors
    t_stat = (estimated_beta - true_beta) / se_beta
    t_statistics.extend(t_stat)

# Compare with t-distribution
df = model.df_resid
x = np.linspace(-4, 4, 100)
t_dist = stats.t(df)

plt.hist(t_statistics, bins=30, density=True, alpha=0.6, color='g', label='t-statistics')
plt.plot(x, t_dist.pdf(x), 'r-', label=f't-distribution (df={int(df)})')
plt.xlabel('t-statistic')
plt.ylabel('Density')
plt.title('T-statistics vs. T-distribution')
plt.legend()
plt.show()
```



3.4 Statistical Properties of LSE

```
##### simulation showing that RSS/sigma^2 indeed follows chi-squared distribution

import numpy as np
import matplotlib.pyplot as plt
import statsmodels.api as sm
from scipy import stats

# Parameters
n_simulations = 500
n_samples = 100
true_beta = np.array([1.5, -2.0, 3.0])
true_error_variance = 1.0
rss_values = []

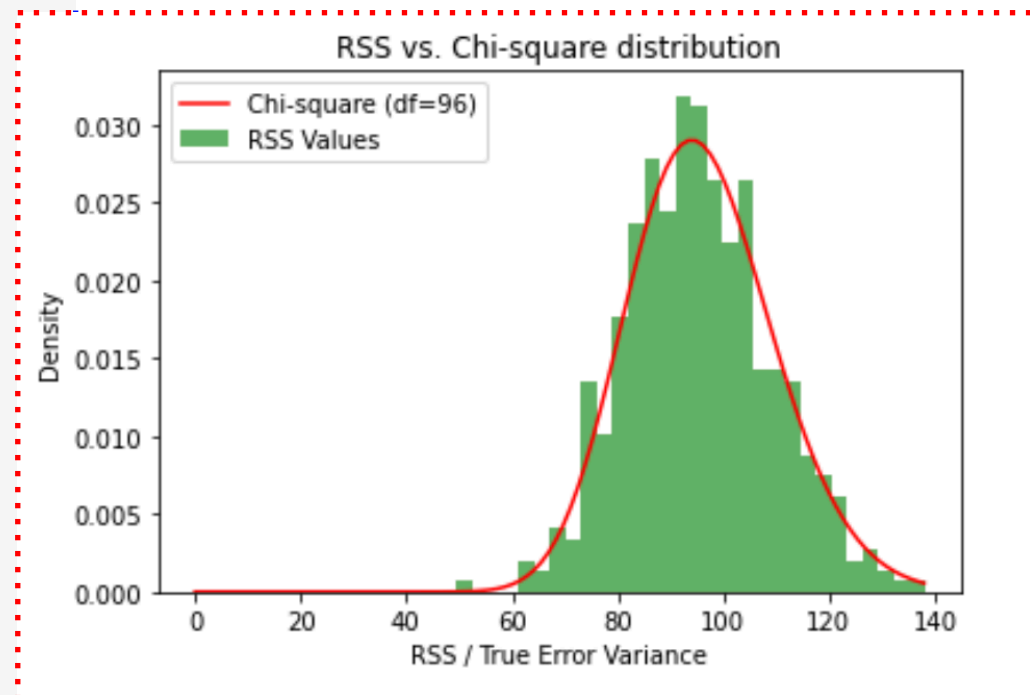
# Simulation loop
np.random.seed(0)
for _ in range(n_simulations):
    X = np.random.normal(size=(n_samples, 3))
    y = X @ true_beta + np.random.normal(scale=np.sqrt(true_error_variance), size=n_samples)

    # Fit the model
    X = sm.add_constant(X)
    model = sm.OLS(y, X).fit()

    # Calculate residual sum of squares
    rss = model.ssr # Sum of squared residuals
    rss_values.append(rss / true_error_variance)

# Compare with chi-square distribution
df_resid = model.df_resid
x = np.linspace(0, max(rss_values), 100)
chi2_dist = stats.chi2(df_resid)

plt.hist(rss_values, bins=30, density=True, alpha=0.6, color='g', label='RSS Values')
plt.plot(x, chi2_dist.pdf(x), 'r-', label=f'Chi-square (df={int(df_resid)})')
plt.xlabel('RSS / True Error Variance')
plt.ylabel('Density')
plt.title('RSS vs. Chi-square distribution')
plt.legend()
plt.show()
```



3.4 Statistical Properties of LSE

$\hat{\beta}$ and $\hat{\epsilon}$ are independent

$$\text{cov}(\hat{\beta}, \hat{\epsilon}) = \text{cov}((\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \epsilon, (\mathbf{I} - \mathbf{H})\epsilon) = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \text{var}(\epsilon)(\mathbf{I} - \mathbf{H}) = \mathbf{0}$$

Uncorrelated normal random variables are independent

Confidence intervals of parameters

$$\hat{\beta}_j \pm t_{n-p-1}(\alpha/2) s \sqrt{c_{jj}}$$

is a confidence interval for β_j at confidence level $1 - \alpha$. Here $t_{n-p-1}(\alpha/2)$ is the $1 - \alpha/2$ percentile of the t -distribution with degree of freedom $n - p - 1$.

$$\mathbb{P}(\beta_j \in [\hat{\beta}_j - t_{n-p-1}(\alpha/2) s \sqrt{c_{jj}}, \hat{\beta}_j + t_{n-p-1}(\alpha/2) s \sqrt{c_{jj}}]) = 1 - \alpha$$

3.4 Statistical Properties of LSE

Confidence intervals of mean response

For a given value of input $\mathbf{x}$ which is a $p + 1$ vector (the first component is constant 1), its mean response is $\beta^T \mathbf{x}$. The confidence interval for this mean response is

$$\hat{\beta}^T \mathbf{x} \pm t_{n-p-1}(\alpha/2)s\sqrt{\mathbf{x}^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}}$$

Prediction interval

To predict the actual response y , rather than its mean, we would use the same point estimator $\hat{\beta}^T \mathbf{x}$, but the accuracy is much decreased as more uncertainty in the randomness of the actual response from the error is involved.

The confidence interval, often called prediction interval, for y is

$$\hat{\beta}^T \mathbf{x} \pm t_{n-p-1}(\alpha/2)s\sqrt{1 + \mathbf{x}^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}}.$$

3.4 Statistical Properties of LSE

Example on Advertising Data by R

```
> ## conf. interval
> mod1<-lm(Sales~TV,data=adv)
> confint(mod1,level=0.95)
```

	2.5 %	97.5 %
(Intercept)	6.12971927	7.93546783
TV	0.04223072	0.05284256

```
> ## pred. interval
> mod1<-lm(Sales~TV,data=adv)
> new_obs <- data.frame(TV=145.3, Radio=51.1, Newspaper=33.9) ## create the new observation x0
> predict(mod1,newdata=new_obs,interval="confidence",level=0.95) ## output the prediction interval of mean response
```

	fit	lwr	upr
1	13.93967	13.48518	14.39416

```
> predict(mod1,newdata=new_obs,interval="prediction",level=0.95) ## output the prediction interval of mean response
```

	fit	lwr	upr
1	13.93967	7.497488	20.38185

3.5 Analysis of Variances (ANOVA)

Decomposition of variances

Recall that $\|\mathbf{y}\|^2 = \|\hat{\mathbf{y}}\|^2 + \|\mathbf{y} - \hat{\mathbf{y}}\|^2$

By subtracting the sample mean $\bar{y}$, then

$$\|\mathbf{y} - \bar{y}\|^2 = \|\hat{\mathbf{y}} - \bar{y}\|^2 + \|\mathbf{y} - \hat{\mathbf{y}}\|^2;$$

which is often written as $\underline{SS_{total} = SS_{reg} + SS_{error}}$.

$$\text{R-squared } R^2 = \frac{\|\hat{\mathbf{y}} - \bar{y}\|^2}{\|\mathbf{y} - \bar{y}\|^2}$$

SS_{total} , the total sum of squares, measures the total variation in response

SS_{reg} , the sum of squares due to regression or, more precisely, due to the inputs, measures variation in response explained by that of the inputs.

SS_{error} , the sum of squares due to error, measures the size of randomness due to error or noise.

3.5 Analysis of Variances (ANOVA)

ANOVA table

Source of Variation	SumOfSquares	Degree of Freedom	Mean Squared	F-statistic
Regression	SS_{reg}	p	MS_{reg}	MS_{reg}/MS_{error}
Error	SS_{error}	$n - p - 1$	MS_{error}	
Total	SS_{total}	$n - 1$		

where $MS_{reg} = SS_{reg}/p$ and $MS_{error} = SS_{error}/(n - p - 1)$.

Is there relation between covariates and response ? (ANOVA)

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_p = 0 \quad \text{v.s.} \quad H_1 : \text{otherwise}$$

And the F -statistic follows $F_{p, n-p-1}$ distribution under the hypothesis that $\beta_1 = \beta_2 = \dots \beta_p = 0$, i.e., all inputs are unrelated with the output.

The p -value is the probability for the distribution $F_{p, n-p-1}$ taking value greater than the value of the F -statistic.

3.5 Analysis of Variances (ANOVA)

Example on Advertising Data by R

To test

$$H_0 : \text{Sales} = \beta_0 + \beta_1 \times \text{TV} + \varepsilon$$

$$H_1 : \text{Sales} = \beta_0 + \beta_1 \times \text{TV} + \beta_2 \times \text{Newspaper} + \beta_3 \times \text{Radio} + \varepsilon$$

```
> ##ANOVA
> mod1<-lm(Sales~TV,data=adv)
> mod2<-lm(Sales~.,data=adv)
> anova(mod1,mod2)
Analysis of Variance Table

Model 1: Sales ~ TV
Model 2: Sales ~ TV + Radio + Newspaper
  Res.Df    RSS Df Sum of Sq    F    Pr(>F)
1     198 2102.53
2     196  556.83  2    1545.7 272.04 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

3.7 Superiority of LSE

Why is LSE good ?

- ▶ *Easy computation*
- ▶ *Consistency* https://en.wikipedia.org/wiki/Consistent_estimator
- ▶ *Efficiency* https://en.wikipedia.org/wiki/Efficient_estimator
- ▶ BLUE (best linear unbiased estimator), among estimates of β , that are linear unbiased estimates: $\sum_{i=1}^n \mathbf{a}_i y_i$, with $\mathbf{a}_i$ being nonrandom.

Theorem (Gauss-Markov Theorem). Among all linear unbiased estimates, the least squares estimate has the smallest variance, thus smallest mean squared error.

3.7 Superiority of LSE

Proof of Gauss-Markov Theorem

- ▶ Let $\mathbf{A} = (\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n)$. Then, unbiased estimate of β is $\mathbf{A}\mathbf{y}$ with mean $\mathbf{A}\mathbf{X}\beta = \beta$ by the unbiasedness, and variance matrix $\mathbf{A}\mathbf{A}^T$.
- ▶ Write $\mathbf{A} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T + \mathbf{D}$. Then, $\mathbf{D}\mathbf{X} = 0$.
- ▶ Then,

$$\mathbf{A}\mathbf{A}^T = (\mathbf{X}^T\mathbf{X})^{-1} + \mathbf{D}\mathbf{D}^T \geq (\mathbf{X}^T\mathbf{X})^{-1}.$$

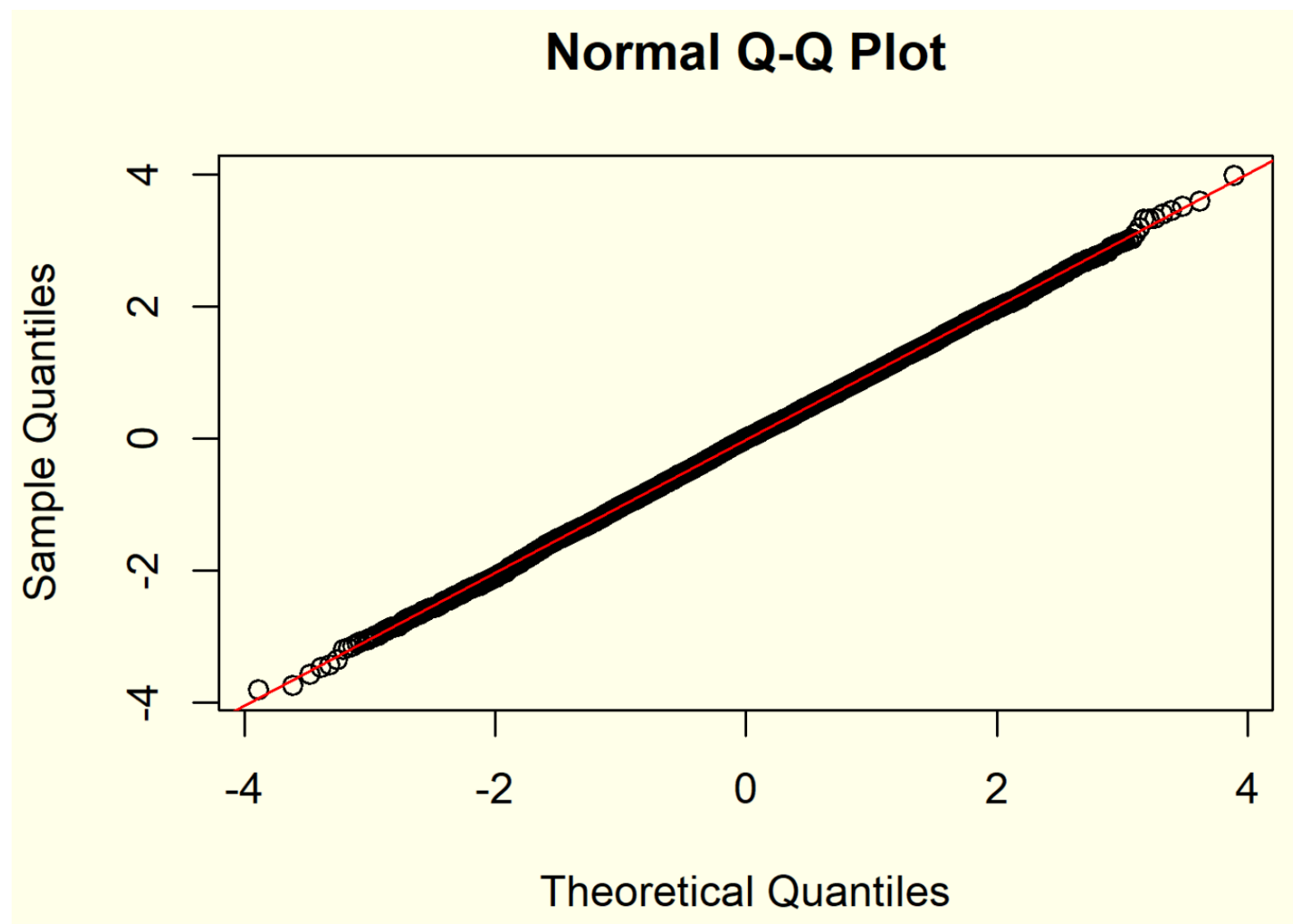
Here the inequality is for symmetric matrices, i.e., $\mathbf{A} \geq \mathbf{B}$ is defined as $\mathbf{A} - \mathbf{B}$ is nonnegative definite.

3.8 Assessing the Regression Model (not required)

a) Error distribution (normality check)

▶ Non-normality may cause the normality-based inference such as *t*-test and *F*-test being inaccurate

▶ Use graphics, such histogram, boxplot and qqnorm to visualize the the distribution of the residuals



ε_i follows normal distributions.

3.8 Assessing the Regression Model (not required)

a) Error distribution (normality check)

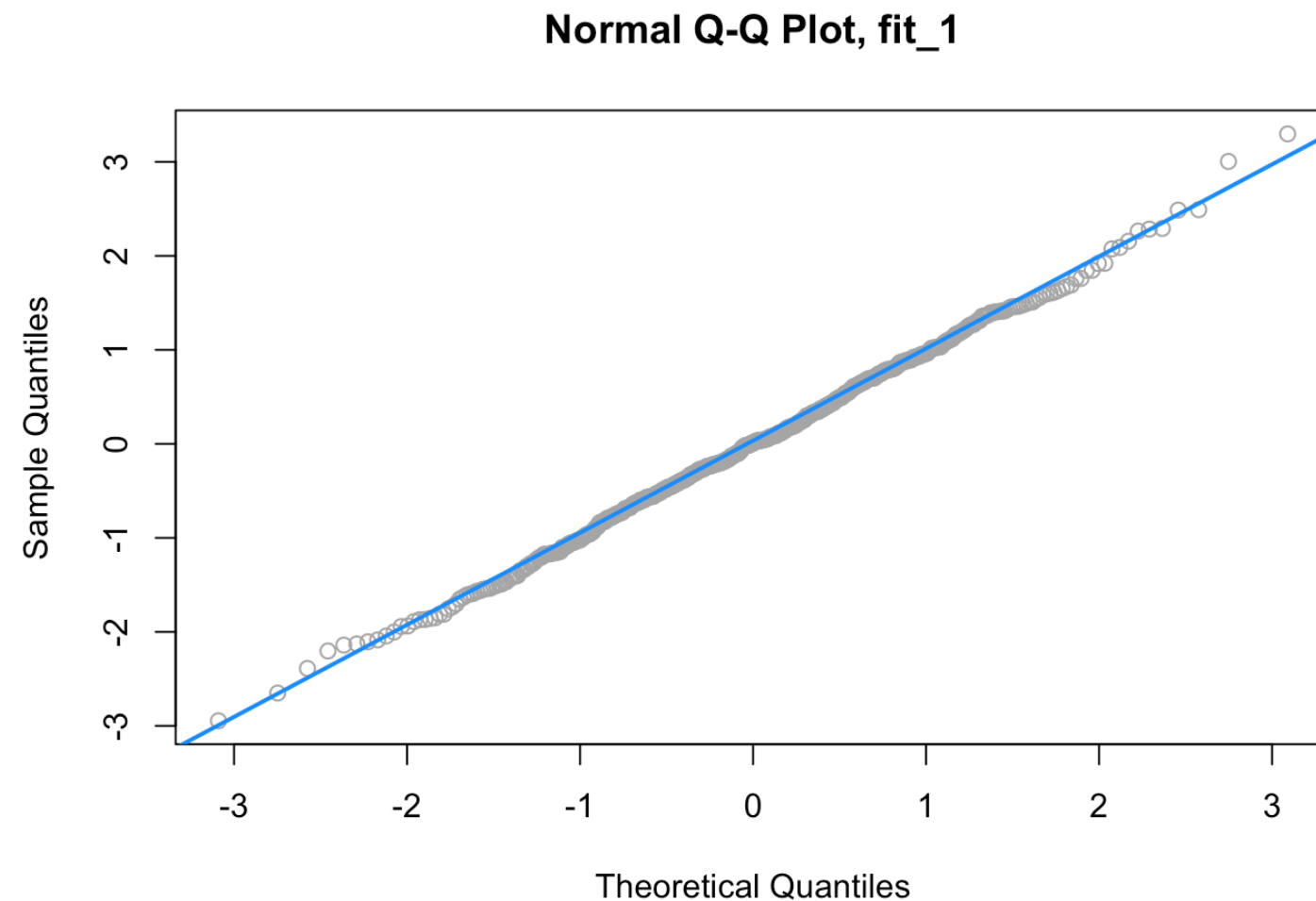
Adjusted residuals

- ▶ the residuals $\hat{\epsilon}_i$ does not follow the distribution $N(0, \sigma^2)$, even if all model assumptions are correct!
- ▶ $\hat{\epsilon} = \mathbf{y} - \hat{\mathbf{y}} = (\mathbf{I} - \mathbf{H})\epsilon \sim N(0, \sigma^2(\mathbf{I} - \mathbf{H}))$. So, $\hat{\epsilon}_i \sim N(0, (1 - h_{ii})\sigma^2)$.
- ▶ *Training error is one of the measurement of the quality of fit.*
- ▶ An (internally) studentized residual is
$$e_i = \frac{\hat{\epsilon}_i}{s\sqrt{1 - h_{ii}}}.$$
- ▶ *A more appropriate one is the (externally) studentized residual which uses an s from the least squares fitting by deleting the i -th observation.*

3.8 Assessing the Regression Model (not required)

a) Error distribution (normality check)

Adjusted residuals



Namely the Q-Q plot but with standardised residuals.

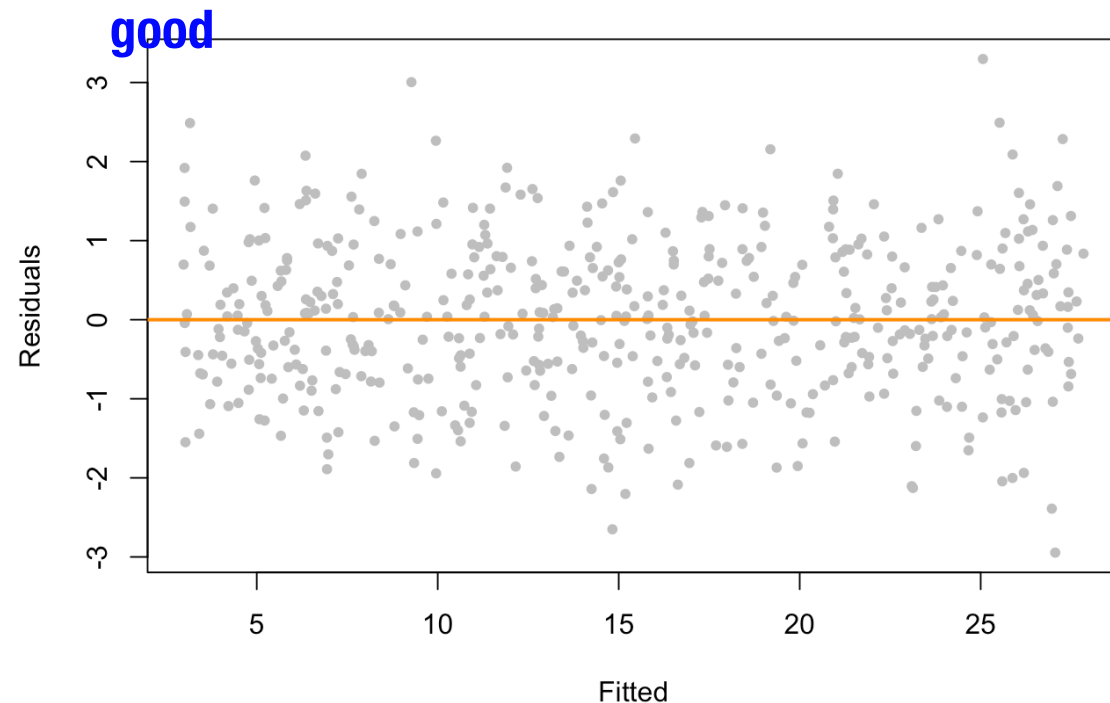
3.8 Assessing the Regression Model (not required)

b) Homoscedasticity versus heteroscedasticity

- ▶ *Heteroscedasticity can cause the estimate being not the optimal one .*
- ▶ *May be fixed by weighted least squares estimation.*
- ▶ *Use scatter plot of residuals against the fitted values to check the heteroscedasticity (the variance of the errors are not equal).*

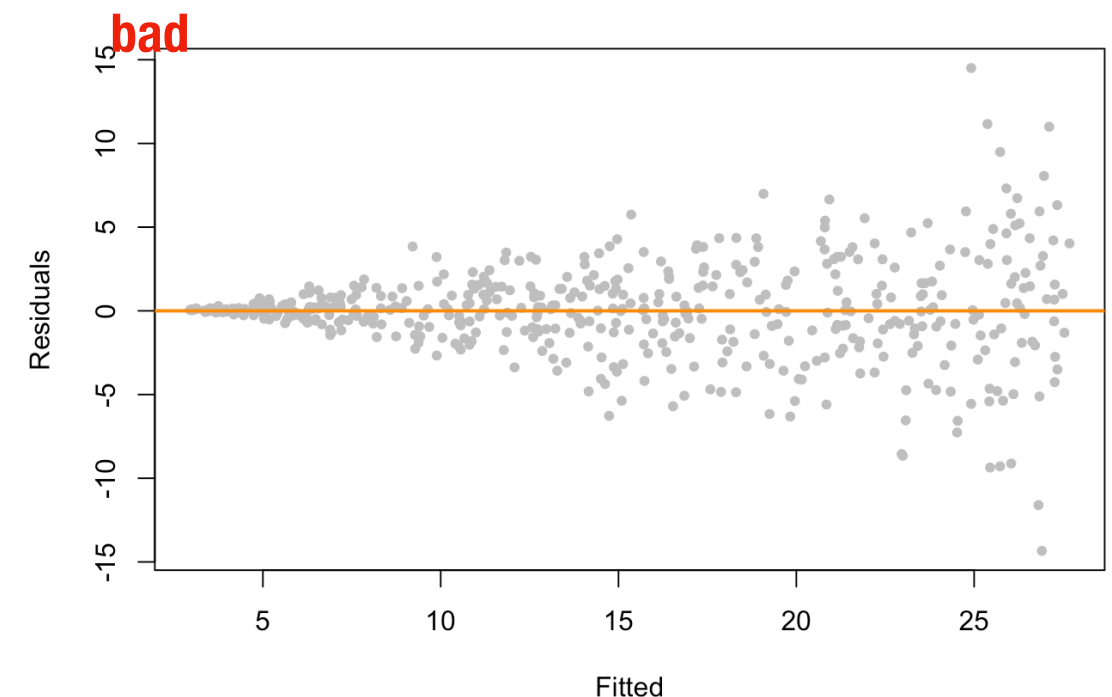
3.8 Assessing the Regression Model (not required)

b) Homoscedasticity versus heteroscedasticity



At any fitted value, the mean of the residuals should be roughly 0. If this is the case, the linearity assumption is valid. For this reason, we generally add a horizontal line at $y=0$ to emphasize this point.

At every fitted value, the spread of the residuals should be roughly the same. If this is the case, the constant variance assumption is valid.

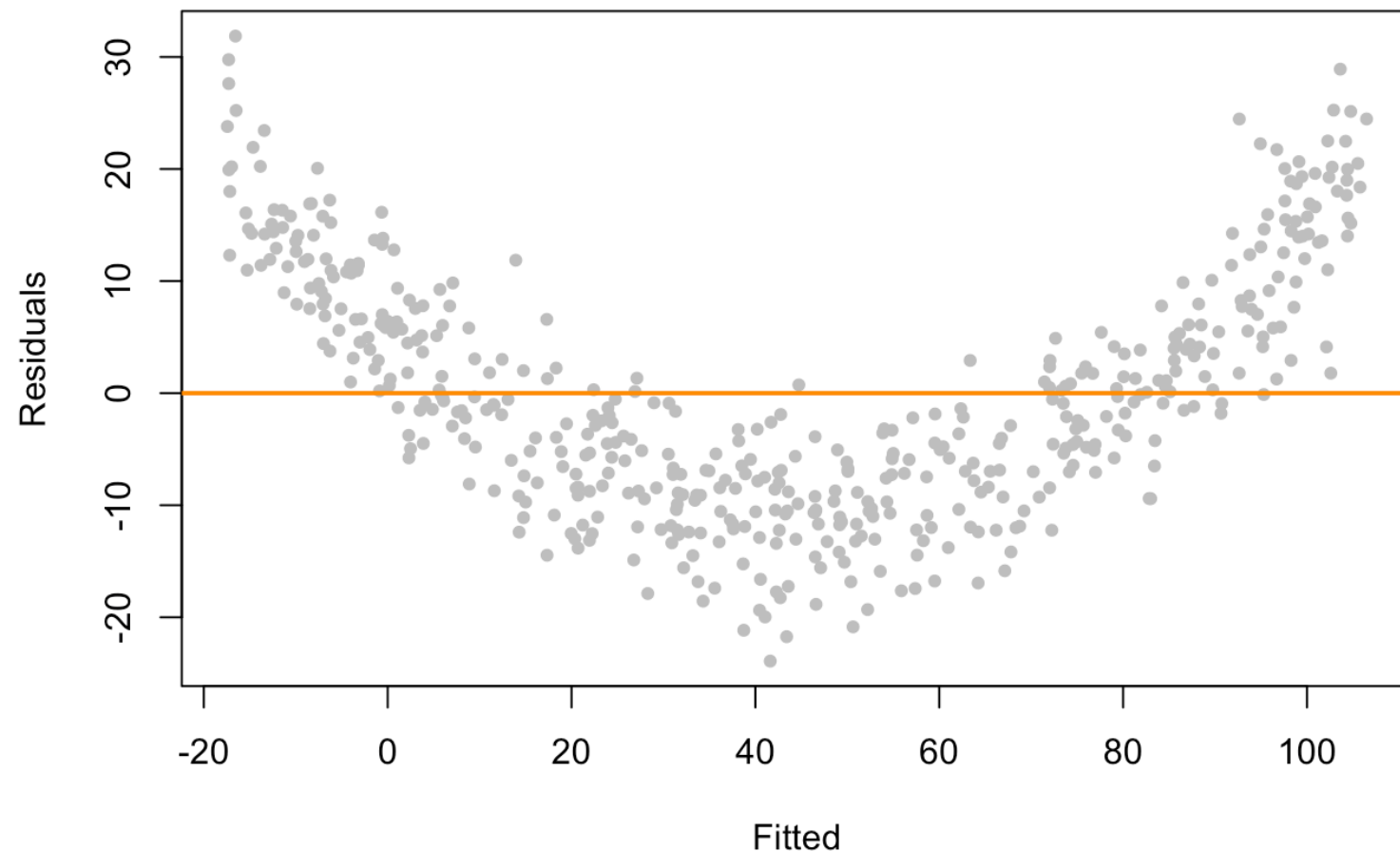


For any fitted value, the residuals seem roughly centered at 0. This is good! The linearity assumption is not violated. However, we also see very clearly, that for larger fitted values, the spread of the residuals is larger. This is bad! The constant variance assumption is violated here.

3.8 Assessing the Regression Model (not required)

b) Homoscedasticity versus heteroscedasticity

Residual versus fitted value



This time on the fitted versus residuals plot, for any fitted value, the spread of the residuals is about the same. However, they are not even close to centered at zero! At small and large fitted values the model is underestimating, while at medium fitted values, the model is overestimating. These are systematic errors, not random noise. So the constant variance assumption is met, but the linearity assumption is violated. The form of our model is simply wrong. We're trying to fit a line to a curve!

3.8 Assessing the Regression Model (not required)

c) Error dependence

▶ The error dependence cause the inference to be incorrect.

▶ Use autocorrelation of the residuals (ACF) to check the independence assumption of the errors.

$$\frac{\sum_{i=2}^n \hat{\epsilon}_i \hat{\epsilon}_{i-1}}{\sqrt{\sum_i \hat{\epsilon}_i^2 \sum_i \hat{\epsilon}_{i-1}^2}}$$

▶ One can also use Durbin-Watson test, which tests whether the first few autocorrelations are 0.

$$d = \frac{\sum_{i=2}^n (\hat{\epsilon}_i - \hat{\epsilon}_{i-1})^2}{\sum_{i=1}^n \hat{\epsilon}_i^2}$$

3.8 Assessing the Regression Model (not required)

c) Error dependence

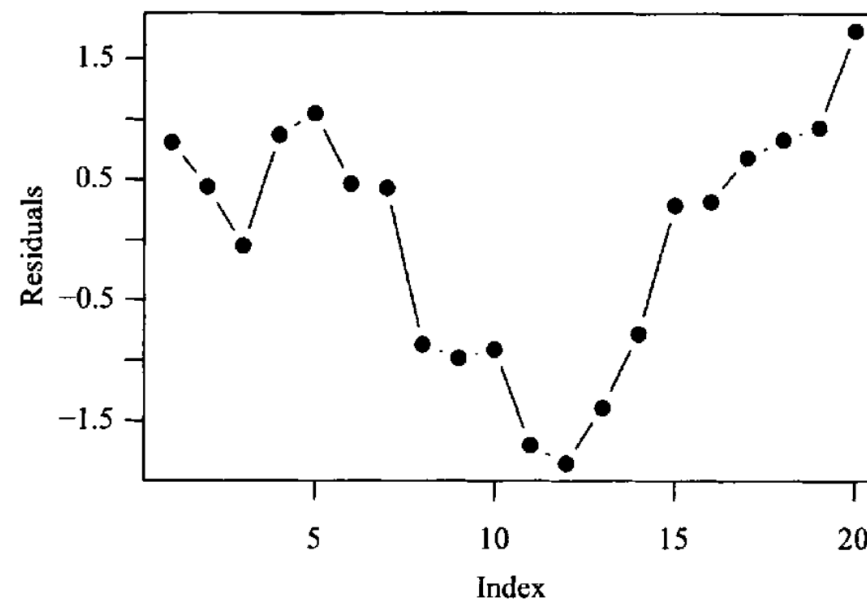


Figure 3.8 Index plot of the standardized residuals.

3.8 Assessing the Regression Model (not required)

d) Leverage and Cook's D

Recall the hat matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T$.

Let $h_{ij} = \mathbf{x}_i^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_j$ be the (i, j) elements of $\mathbf{H}$.

The leverage of the i -th observation is just the i -th diagonal element of $\mathbf{H}$, denoted as h_{ii} .

A high leverage implies that observation is quite influential. Note that the average of h_{ii} is $(p + 1)/n$.

Leverage is to detect outliers in X-space

So, if h_{ii} is greater than $2(p + 1)/n$, twice of the average, is generally considered large.

3.8 Assessing the Regression Model (not required)

d) Leverage and Cook's D

▶ The Cook's D is often used to measure how important an observation is. Cook's D is defined

$$D_i = \frac{\sum_{k=1}^n (\hat{y}_k - \hat{y}_k^{(-i)})^2}{(p+1)s^2}$$

where $\hat{y}_k$ is the k -th fitted value; and $\hat{y}_k^{(-i)}$ is the k -th fitted value by deleting the i -th observation.

▶ If D_i is large, it implies once i -th observation is not available, the prediction would be much different, thus reflecting the importance of this observation.

Cook's D is to measure the influential of a data point

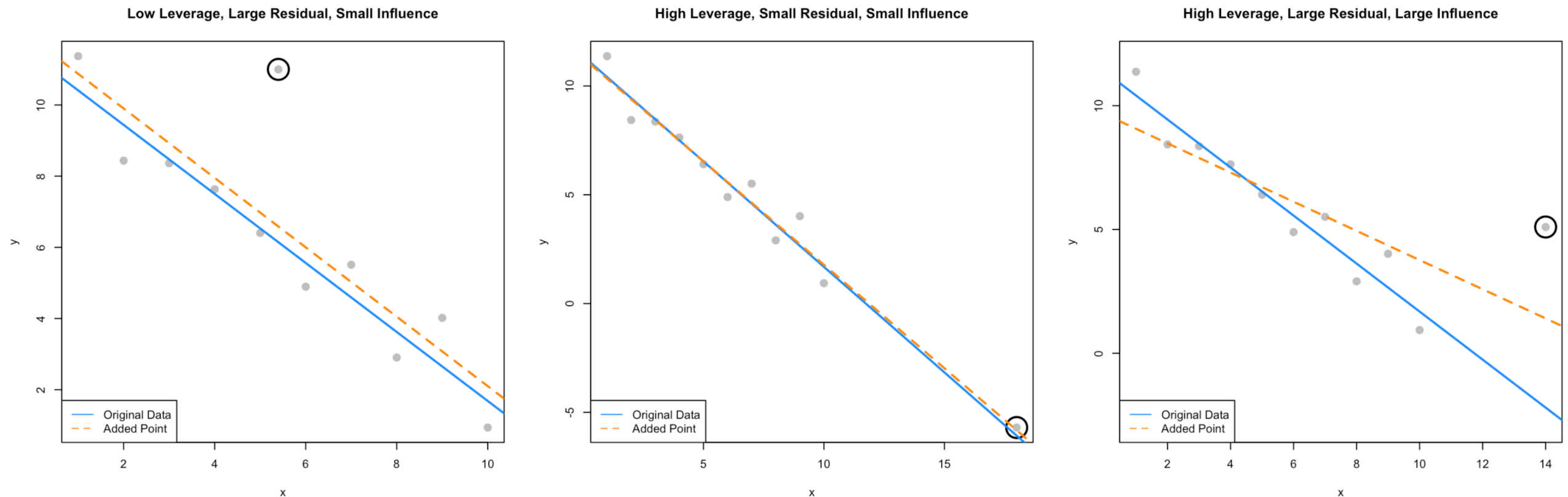
▶ In general, the observations with large D_i , such as larger than a quarter of the sample size, may be considered influential.

3.8 Assessing the Regression Model (not required)

d) Leverage and Cook's D

These three examples illustrate the relationship between **Cook's distance**, **residual** and **leverage**.

For simple linear regression, x_i is far away from $\bar{x}$, then it has high leverage



The blue line is obtained without using the circled point; the orange line is obtained including the circled point

3.9 Variable Selection

Model 1

use all p predictors

$$\mathbf{X} = \begin{pmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1p} \\ 1 & x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{np} \end{pmatrix} \quad \longrightarrow \quad \mathbf{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$$

Model 2

use only q predictors
 $q < p$

$$\mathbf{X} = \begin{pmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1,q} \\ 1 & x_{21} & x_{22} & \cdots & x_{2,q} \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{n,q} \end{pmatrix} \quad \longrightarrow \quad \mathbf{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$$

How should we assess which model is better ?

3.9 Variable Selection

- ▶ *Variable selection, or more generally, model selection, is an important tool in minimizing prediction error.*
- ▶ *There are substantial research development regarding methods of model selection.*
- ▶ *The aim is to minimize generalization error or expected error.*
- ▶ *The naive approach is to exhaust all models. However, with the curse of dimensionality, this is quickly prohibitive when the number of variables increase.*
- ▶ *Variable More sophisticated methods such as cross validation or regularization methods, such as LASSO (Tibshirani 1996).*

3.9 Variable Selection

▶ *More inputs do not imply better prediction, particularly if the inputs in the model are irrelevant with the response.*

▶ *Moreover, more inputs also imply more danger of overfit, resulting in small training error but large test error.*

▶ *Here we introduce more basic and simple methods.*

3.9 Variable Selection

a) Adjusted R-squared

▶ *The R-squared is the percentage of the total variation in response due to the inputs. The R-squared is commonly used as a measurement of how good the linear fit is.*

▶ *However, a model with larger R-squared is not necessarily better than another model with smaller R-squared!*

▶ *If model A has all the inputs of model B, then model A's R-squared will always be greater than or as large as that of model B.*

▶ *If model A's additional inputs are entirely uncorrelated with the response, model A contain more noise than model B. As a result, the prediction based on model A would inevitably be poorer or no better.*

3.9 Variable Selection

a) Adjusted R-squared

Recall that the R-squared is defined as: $R^2 = \frac{SS_{reg}}{SS_{total}} = 1 - \frac{SS_{error}}{SS_{total}}$

where $SS_{error} = \sum_{i=1}^n \hat{\epsilon}_i^2$ is often called residual sum of squares (RSS).

The adjusted R-squared, taking into account of the degrees of freedom, is defined as

$$\text{adjusted } R^2 = 1 - \frac{MS_{error}}{MS_{total}} = 1 - \frac{SS_{error}/(n - p - 1)}{SS_{total}/(n - 1)} = 1 - \frac{s^2}{\sum_{i=1}^n (y_i - \bar{y})^2 / (n - 1)}$$

With more inputs, the R-squared always increase, but the adjusted R-squared could decrease since more inputs is penalized by the smaller degree of freedom of the residuals.

The adjusted R-squared is preferred over the R-squared in evaluating models.

3.9 Variable Selection

b). Mallows' C_p

Recall that our linear model has p covariates, and

$s^2 = SS_{error}/(n - p - 1)$ is the unbiased estimator of σ^2 .

Assume now more covariates are available. Suppose we use only p of the K covariates with $K \geq p$.

The statistic of Mallows' C_p is defined as

$$\frac{SS_{error}(p)}{s_K^2} + 2(p + 1) - n.$$

where SS_{error} is the residual sum of squares for the linear model with p inputs and s_K^2 is the unbiased estimator of σ^2 based on K inputs.

The smaller Mallows' C_p is, the better the model is.

The following AIC is more often used, despite that Mallows' C_p and AIC usually give the same best model.

3.9 Variable Selection

c) AIC

AIC stands for Akaike information criterion, which is defined as

$$AIC = \log(s^2) + 2(1 + p)/n,$$

for a linear model with p inputs, where $s^2 = SS_{error}/(n - p - 1)$.

AIC aims at maximizing the predictive likelihood. The model with the smallest AIC is preferred.

The AIC criterion is try to maximize the expected predictive likelihood. In general, it can be roughly derived in the following. Let θ be a parameter of d dimension. $\hat{\theta}$ is the maximum likelihood estimator of θ based on observations $y_1, \dots, y_n$. Let θ_0 be the true (unknown) value of θ , and $\mathcal{I}(\theta_0)$ be the Fisher information.

3.9 Variable Selection

c) AIC

The (expected) predictive log-likelihood

$$\begin{aligned}
 & E(\log f(Y|\theta))|_{\theta=\hat{\theta}} \\
 \approx & E(\log f(Y|\theta))|_{\theta=\theta_0} - \frac{1}{2}(\hat{\theta} - \theta_0)^T \mathcal{I}(\theta_0)(\hat{\theta} - \theta_0) \\
 \approx & \frac{1}{n} \sum_{i=1}^n \log f(y_i|\hat{\theta}) - \frac{1}{2}(\hat{\theta} - \theta_0)^T \mathcal{I}(\theta_0)(\hat{\theta} - \theta_0) \\
 & - \frac{1}{2}(\hat{\theta} - \theta_0)^T \mathcal{I}(\theta_0)(\hat{\theta} - \theta_0) \\
 \approx & \frac{1}{n}(\text{maximum log likelihood}) - (\hat{\theta} - \theta_0)^T \mathcal{I}(\theta_0)(\hat{\theta} - \theta_0) \\
 \approx & \frac{1}{n}(\text{maximum log likelihood}) - d/n
 \end{aligned}$$

The approximations are due to the Taylor expansion.

3.9 Variable Selection

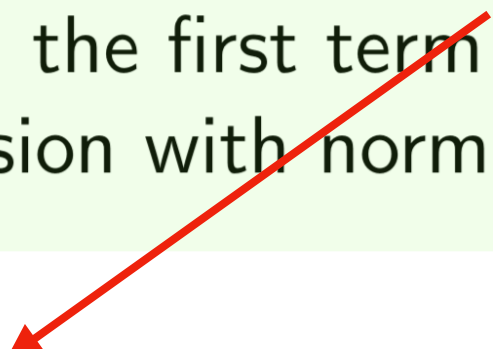
c) AIC

The (expected) predictive log-likelihood

Then, maximizing the above predictive likelihood is the same as minimize

$$\underline{-2(\text{maximum log likelihood}) + 2d}$$

where, the first term is called deviance. In the case of linear regression with normal errors, the deviance is the same as $\log(s^2)$.



The formula of AIC in general models.

In the special case of linear regression model, it reduces to $\text{AIC} = \log(s^2) + 2(1 + p)/n$

3.9 Variable Selection

d) BIC

BIC stands for Schwarz's Bayesian information criterion, which is defined as


$$\text{BIC} = \log(s^2) + (1 + p) \log(n)/n,$$

for a linear model with p inputs. Again, the model with the smallest BIC is preferred.



The derivation of BIC results from Bayesian statistics and has Bayesian interpretation. It is seen that BIC is formally similar to AIC. The BIC penalizes more heavily the models with more number of inputs.

3.9 Variable Selection

```
##### Multiple linear regression, only X1 and X2 are useful, report the AIC, BIC, Mallow's Cp and Adjusted Rsqu

import numpy as np
import pandas as pd
import statsmodels.api as sm
from itertools import combinations

# Generate synthetic data
np.random.seed(0)
n_samples = 100
X = np.random.normal(size=(n_samples, 4))
true_beta = np.array([2.0, -1.5, 0, 0]) # Only first 2 predictors are useful
y = X @ true_beta + np.random.normal(scale=1.0, size=n_samples)

# DataFrame for predictors
df = pd.DataFrame(X, columns=[f'X{i+1}' for i in range(X.shape[1])])

# Function to calculate Mallow's Cp
def mallows_cp(model, sigma_squared):
    p = model.df_model + 1
    rss = model.ssr
    return (rss / sigma_squared) - (len(y) - 2 * p)

# Fit models of different sizes
results = []

for k in range(1, 5):
    for combo in combinations(df.columns, k):
        X_subset = sm.add_constant(df[list(combo)])
        model = sm.OLS(y, X_subset).fit()

        # Calculate statistics
        aic = model.aic
        bic = model.bic
        adj_r_squared = model.rsquared_adj
        sigma_squared = model.mse_resid
        cp = mallows_cp(model, sigma_squared)

        results.append({
            'Predictors': combo,
            'AIC': aic,
            'BIC': bic,
            'Mallow's Cp': cp,
            'Adjusted R^2': adj_r_squared
        })

# Convert results to DataFrame
results_df = pd.DataFrame(results)
print(results_df.sort_values('AIC').reset_index(drop=True))
```

	Predictors	AIC	BIC	Mallow's Cp	Adjusted R ²
0	(X1, X2)	293.641130	301.456640	3.0	0.868893
1	(X1, X2, X3)	295.553610	305.974291	4.0	0.867643
2	(X1, X2, X4)	295.615241	306.035922	4.0	0.867561
3	(X1, X2, X3, X4)	297.527426	310.553277	5.0	0.866285
4	(X1, X3)	412.313657	420.129168	3.0	0.570449
5	(X1, X3, X4)	413.958476	424.379157	4.0	0.567513
6	(X1,)	414.311089	419.521429	2.0	0.557492
7	(X1, X4)	415.944919	423.760429	3.0	0.554564
8	(X2,)	450.163646	455.373987	2.0	0.366675
9	(X2, X3)	450.501357	458.316868	3.0	0.370694
10	(X2, X4)	451.795263	459.610773	3.0	0.362499
11	(X2, X3, X4)	452.141374	462.562055	4.0	0.366424
12	(X3,)	496.794066	502.004407	2.0	-0.009578
13	(X4,)	496.836995	502.047335	2.0	-0.010012
14	(X3, X4)	498.773571	506.589081	3.0	-0.019777

3.10 Comparison with KNN Regression

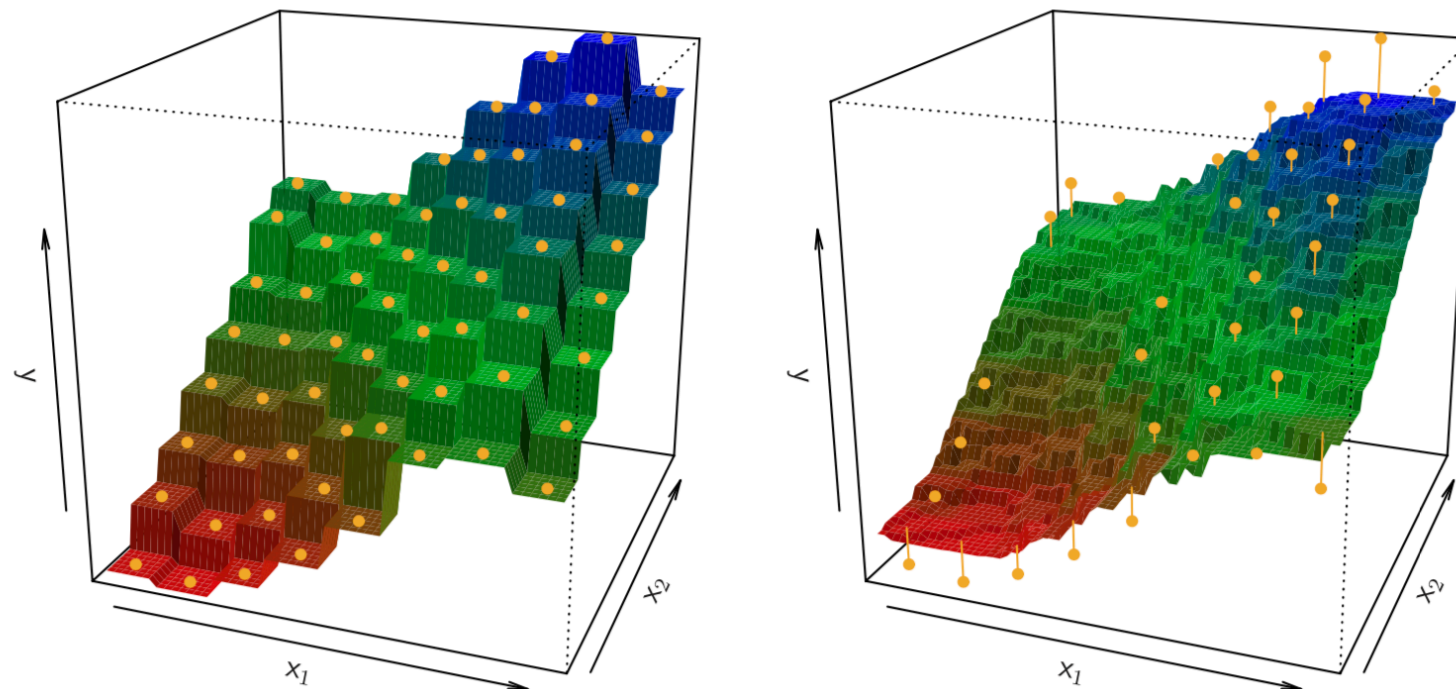
KNN regression

Given data $(x_1, y_1), \dots, (x_n, y_n)$, KNN predicts $f(x_0)$ by

$$\hat{f}(x_0) = K^{-1} \sum_{i \in \mathcal{N}_K(x_0)} y_i$$

where $\mathcal{N}_K(x_0)$ contains the top-K data points closest to x_0

Example 1



The KNN regression on a two-dimensional data set with 64 observations (orange dots). Left: $K = 1$ results in a rough step function fit. Right: $K = 9$ produces a much smoother fit.

3.10 Comparison with KNN Regression

Example 2

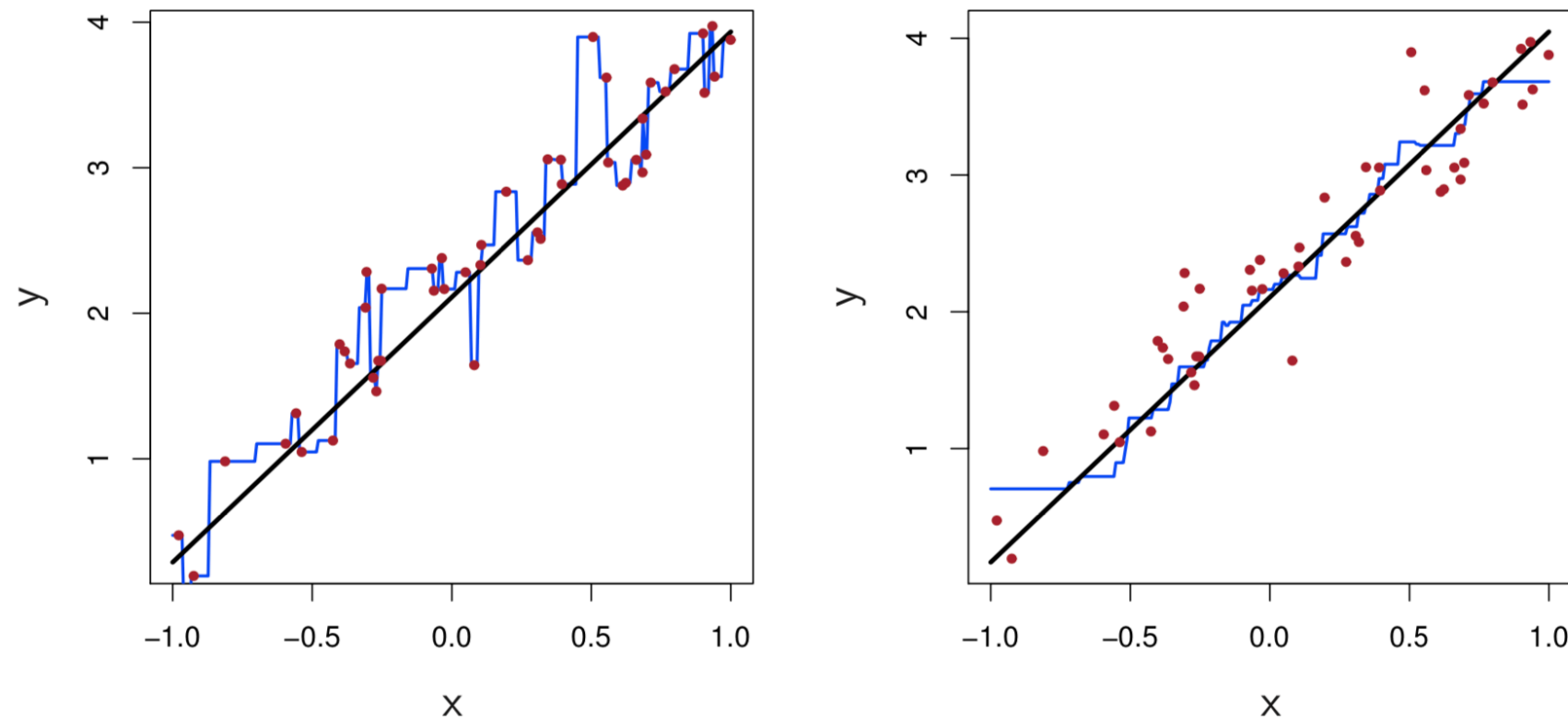


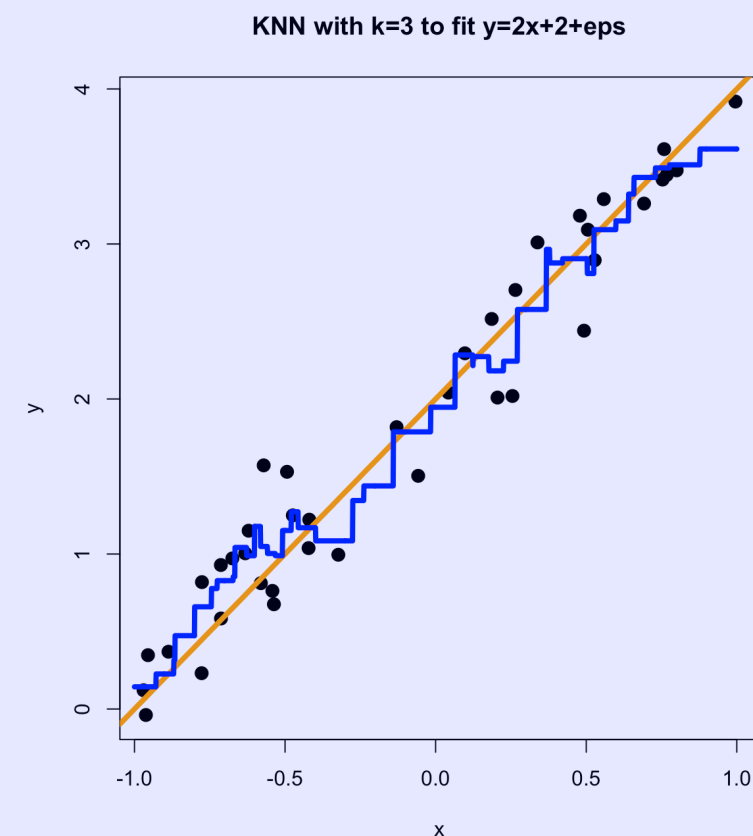
Figure: 3.17. Plots of $\hat{f}(X)$ using KNN regression on a one-dimensional data set with 100 observations. The true relationship is given by the black solid line. Left: The blue curve corresponds to $K = 1$ and interpolates (i.e. passes directly through) the training data. Right: The blue curve corresponds to $K = 9$, and represents a smoother fit.

3.10 Comparison with KNN Regression

Example 2 by R

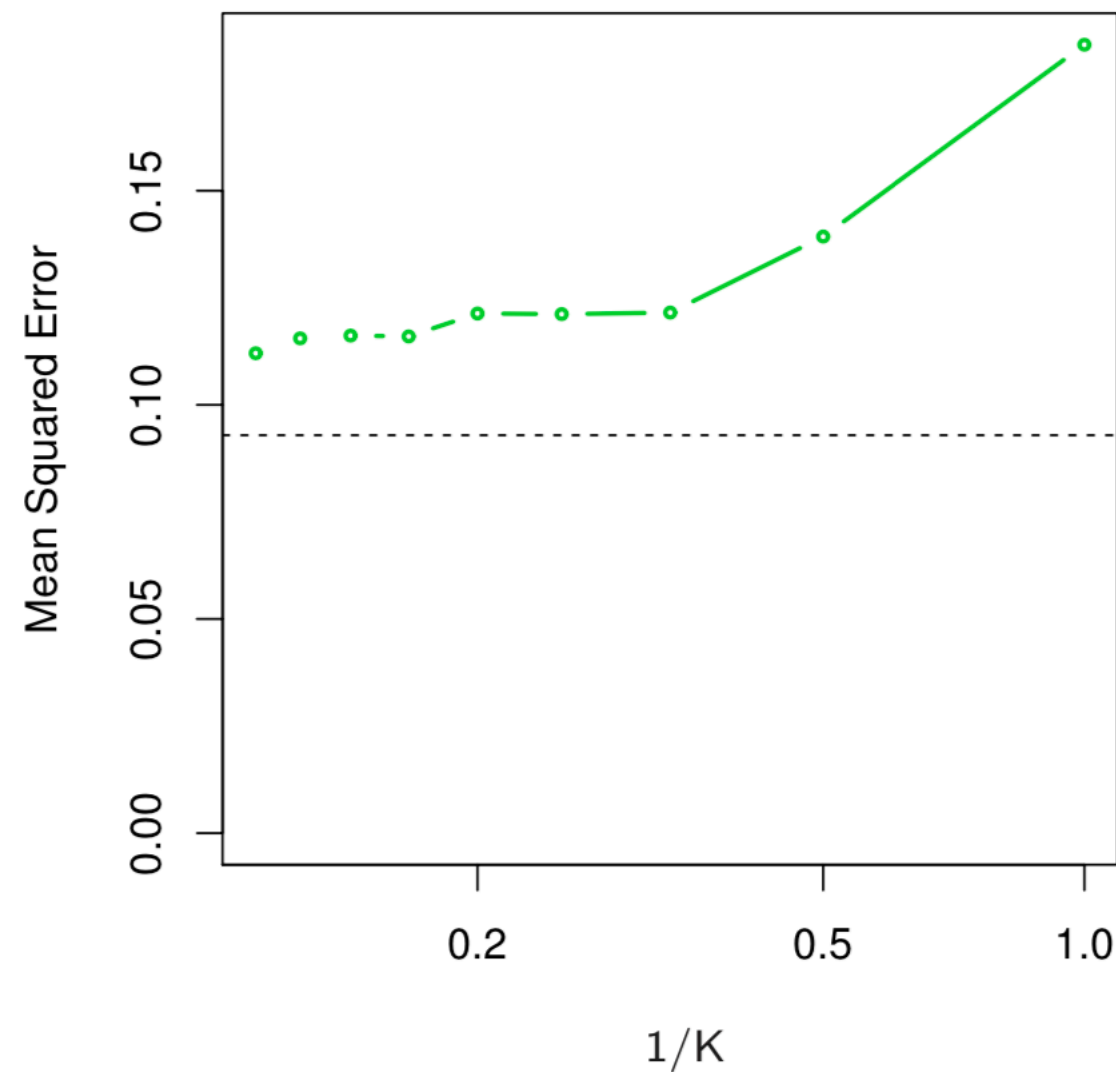
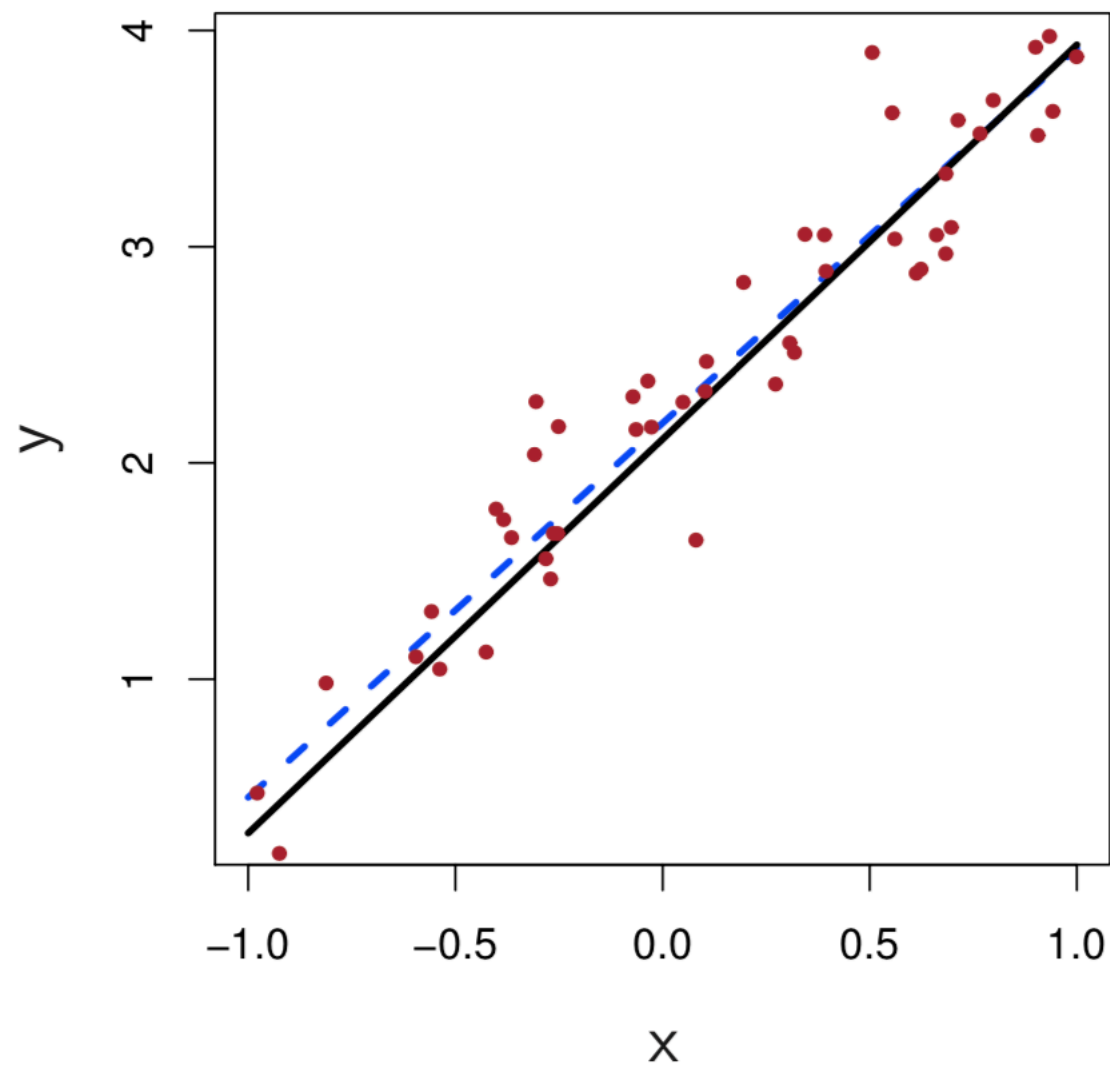
```
> ##### KNN regression
> library(caret)
> train_x<-data.frame(x=runif(40,min=-1,max=1))
> train_y<-2*train_x$x+2+rnorm(40,mean=0,sd=0.3)
> knnmod1 <- knnreg(train_x, train_y,k=3)
> str(knnmod1)
List of 3
 $ learn :List of 2
  ..$ y: num [1:40] 0.676 3.092 0.812 0.972 1.222 ...
  ..$ X:'data.frame': 40 obs. of 1 variable:
    .. ..$ x: num [1:40] -0.537 0.505 -0.58 -0.675 -0.42 ...
 $ k      : num 3
 $ theDots: list()
- attr(*, "class")= chr "knnreg"
```

```
## plot
test_x<-data.frame(x=seq(-1,1,0.001))
test_y=predict(knnmod1,test_x)
plot(train_x$x,train_y,pch = 20,
      cex = 2,
      col = "black",
      main="KNN with k=3 to fit y=2x+2+eps",
      xlab="x",
      ylab="y")
abline(2,2,col="orange",lwd=4)
lines(test_x$x,test_y,col="blue",lwd=4)
```



3.10 Comparison with KNN Regression

Example 3



Details on next slide

3.10 Comparison with KNN Regression

Example 3 (cont.)

The same data set shown in Example 2 is investigated further.

Left: *The blue dashed line is the least squares fit to the data. Since $f(X)$ is in fact linear (displayed as the black line), the least squares regression line provides a very good estimate of $f(X)$.*

Right: *The dashed horizontal line represents the least squares test set MSE, while the green solid line corresponds to the MSE for KNN as a function of $1/K$ (on the log scale). Linear regression achieves a lower test MSE than does KNN regression, since $f(X)$ is in fact linear. For KNN regression, the best results occur with a very large value of K , corresponding to a small value of $1/K$.*

3.10 Comparison with KNN Regression

Example 4

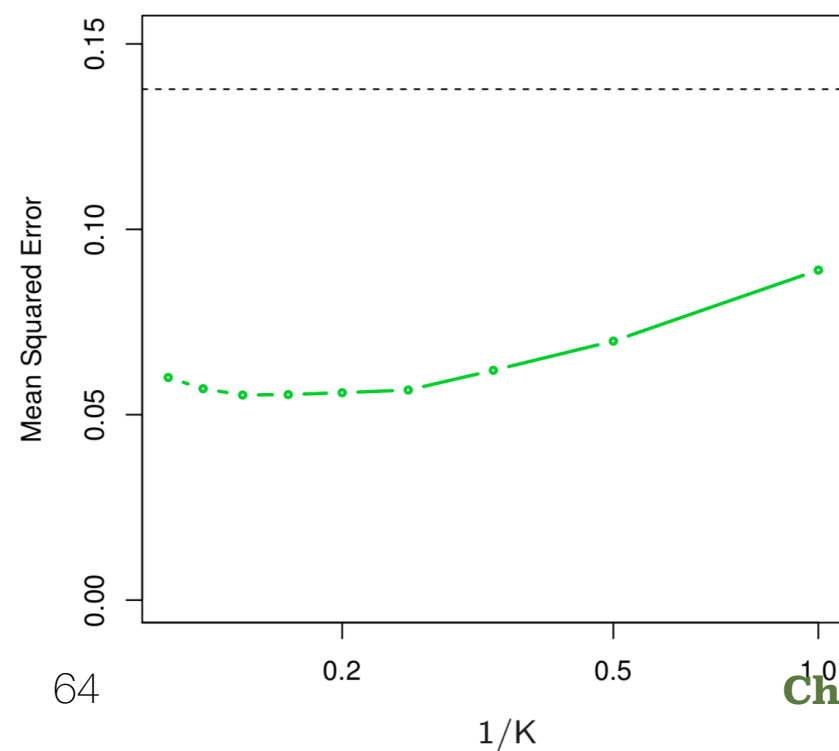
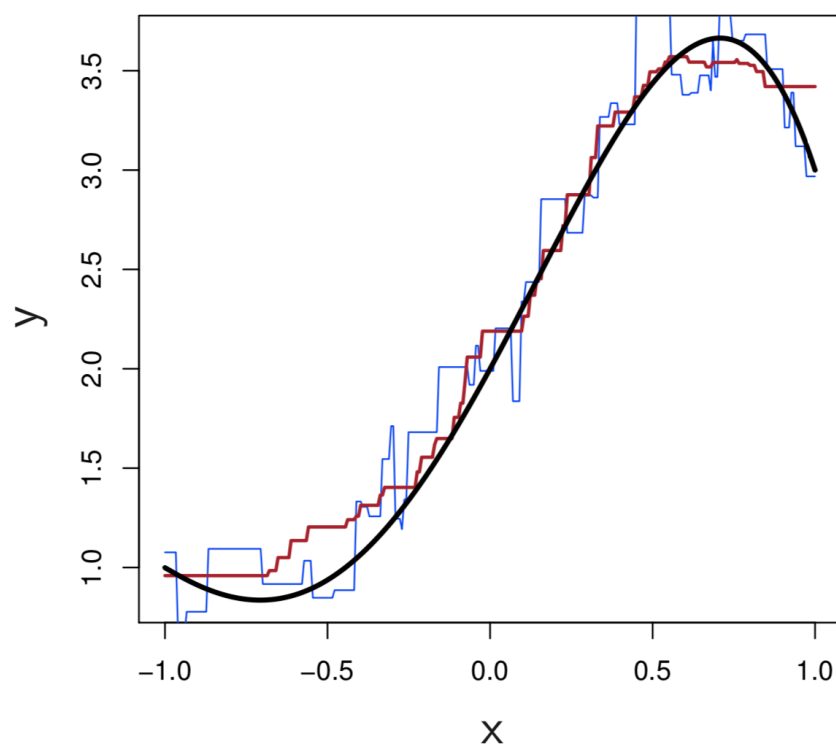
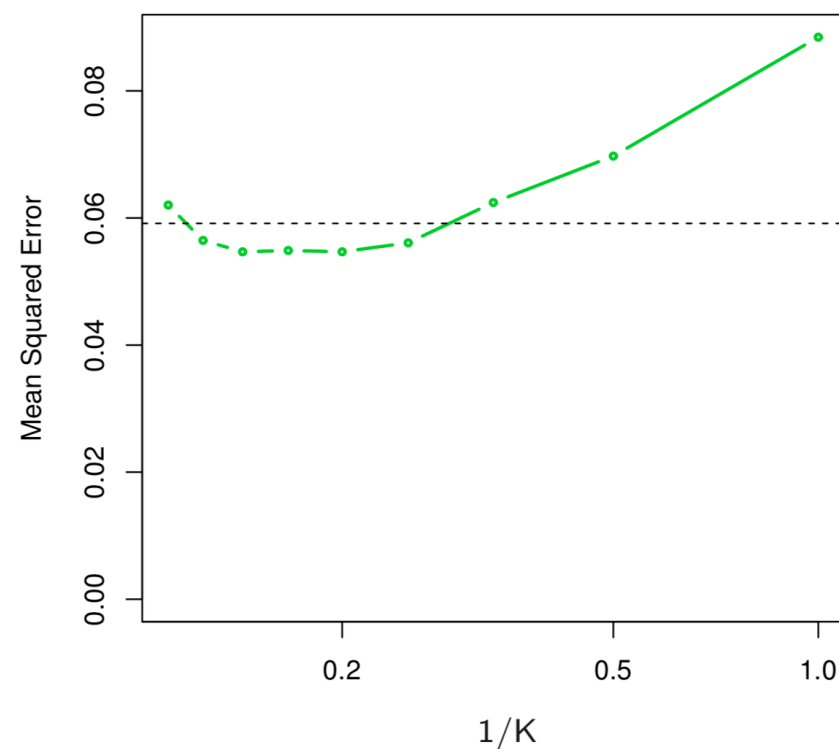
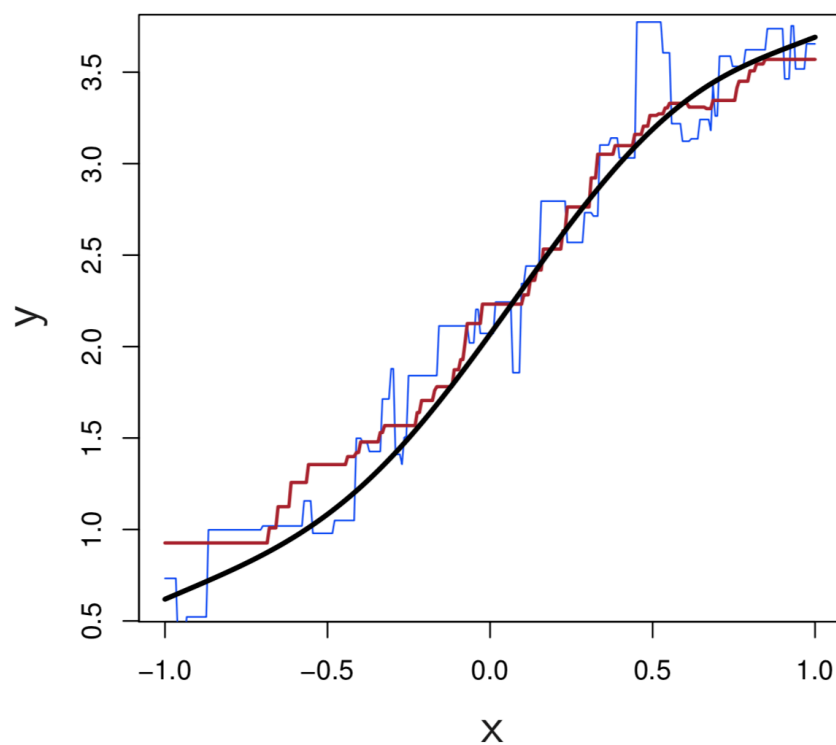


Figure 3.19

➡
Details on next slide

3.10 Comparison with KNN Regression

Example 4 (cont.)

Top Left: In a setting with a slightly non-linear relationship between X and Y (solid black line), the KNN fits with $K = 1$ (blue) and $K = 9$ (red) are displayed.

Top Right: For the slightly non-linear data, the test set MSE for least squares regression (horizontal black) and KNN with various values of $1/K$ (green) are displayed.

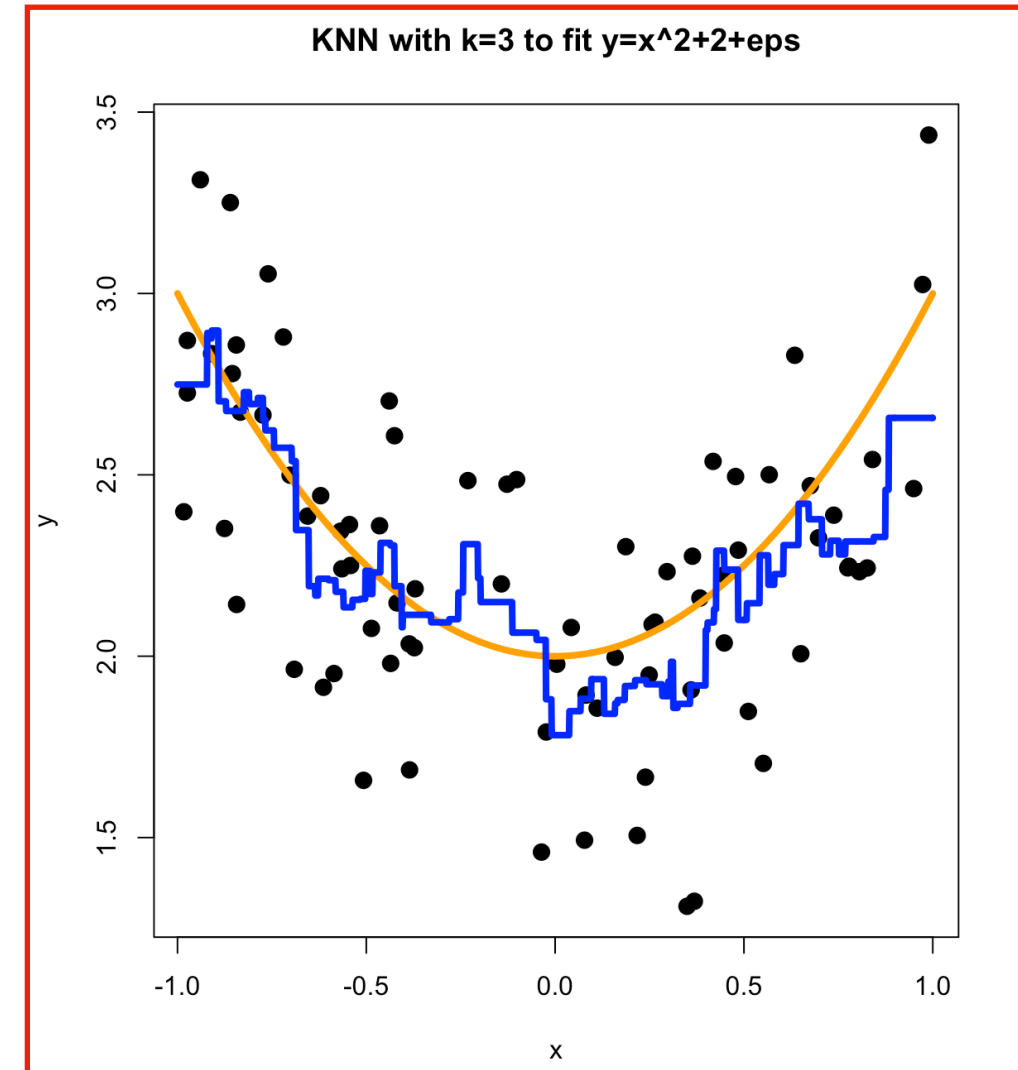
The **Bottom Left** and **Bottom Right:** As in the top panel, but with a strongly non-linear relationship between X and Y .

3.10 Comparison with KNN Regression

Example 4 by R

```
##### KNN regression
library(caret)
train_x<-data.frame(x=runif(80,min=-1,max=1))
train_y<-(train_x$x)^2+2+rnorm(40,mean=0,sd=0.3)
knnmod2 <- knnreg(train_x, train_y,k=6)
str(knnmod2)

## plot
test_x<-data.frame(x=seq(-1,1,0.001))
test_y=predict(knnmod2,test_x)
plot(train_x$x,train_y,pch = 20,
      cex = 2,
      col = "black",
      main="KNN with k=3 to fit y=x^2+2+eps",
      xlab="x",
      ylab="y")
lines(test_x$x,(test_x$x^2)+2,col="orange",lwd=4)
lines(test_x$x,test_y,col="blue",lwd=4)
```



3.10 Comparison with KNN Regression

Example 5

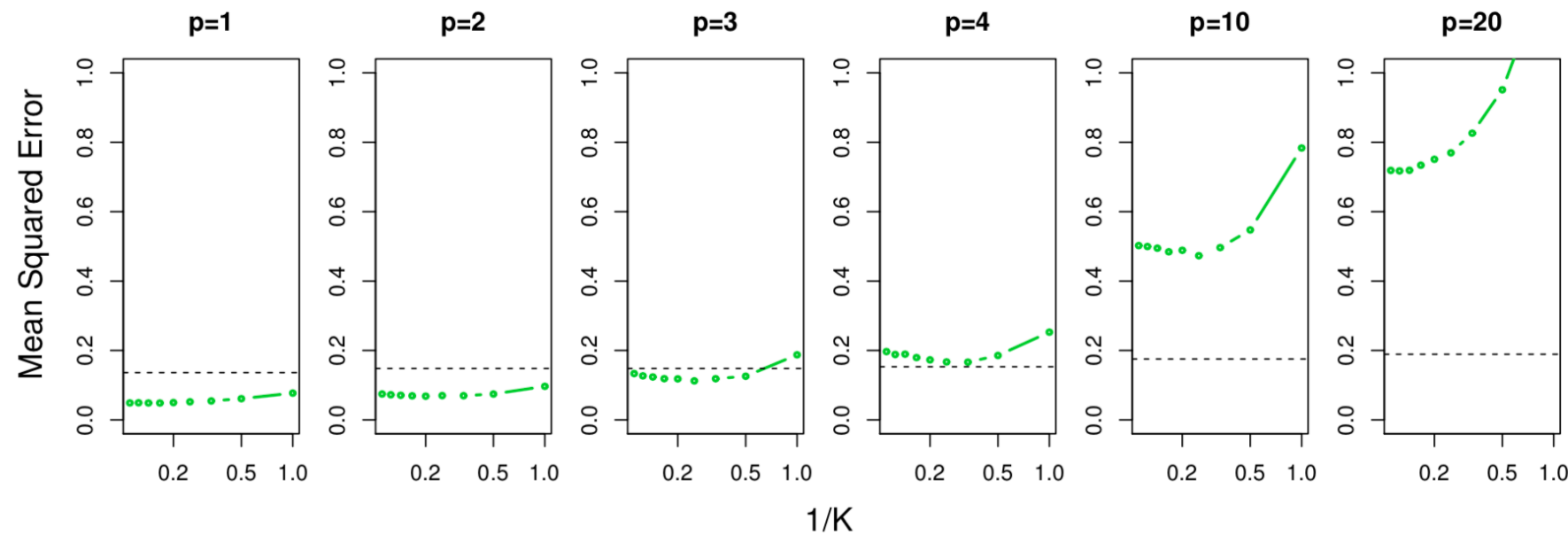


Figure: 3.20. Test MSE for linear regression (black dashed lines) and KNN (green curves) as the number of variables p increases. The true function is non linear in the first variable, as in the lower panel in Figure 3.19, and does not depend on the additional variables. The performance of linear regression deteriorates slowly in the presence of these additional noise variables, whereas KNNs performance degrades much more quickly as p increases.